

GlucoSnap Product Requirements Document

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Executive Summary

GlucoSnap is a mobile-first prediabetes reversal companion application that transforms the overwhelming experience of managing prediabetes into a confident, actionable daily practice. By combining instant photo-based meal analysis with glycemic load tracking and personalized guidance, GlucoSnap addresses the critical gap between diagnosis and successful behavior change.

The Problem

88 million Americans have prediabetes, representing an \$8.81 billion market expected to reach \$15.1 billion by 2032[1][2]. Despite this massive scale, the overwhelming majority of prediabetic individuals face the same crisis immediately after diagnosis: complete nutritional paralysis. As one Reddit user captured perfectly: "THERE IS NO FOOD LEFT. HOW DO YOU ALL LIVE."[3]

The core unmet need is not information—it is actionable, personalized, in-the-moment clarity about food choices. Existing applications focus on Type 1/2 diabetics, track generic calories, and require extensive manual data entry. None answer the fundamental question every prediabetic asks 3-5 times daily: "**Can I eat this? Will this spike me?**"

The Solution

GlucoSnap delivers instant clarity through photo-based meal scanning specifically designed for prediabetes reversal. Users point their phone camera at any meal and receive within 5 seconds:

- Glycemic Load (GL) score - the #1 blood sugar predictor[4][5]
- Spike risk rating (Safe/Moderate/High)
- Food sequencing advice (eat this first, this last)[6][7]
- Safer food swaps maintaining taste and satisfaction
- Post-meal action recommendations (walk timing and duration)
- Real-time daily GL budget tracking
- 90-day A1C reversal progress visualization

Core Differentiators

Feature	GlucoSnap	Cal AI	mySugr	GlycoAI
Prediabetes-exclusive focus	✓	✗	✗	✗
Photo-first meal scanning	✓	✓	✗	✓
Glycemic Load tracking	✓	✗	✗	Partial
Food sequencing guidance	✓	✗	✗	✗
A1C reversal roadmap	✓	✗	✗	✗
Zero manual entry	✓	✓	✗	Partial

Table 1: Competitive feature comparison matrix

Market Opportunity

- **Target market:** 88 million prediabetics in the US alone[1]
- **Global market:** \$8.81B (2025) growing to \$15.1B (2032)[2]
- **Digital health diabetes apps:** \$61.2B market by 2030[8]
- **Comparable success:** Cal AI achieved \$2M MRR from photo-based food scanning with a general audience[9]

Success Metrics - 12 Month Targets

- **Downloads:** 25,000 total installations
- **DAU/MAU ratio:** 45% (industry benchmark: 20%)
- **Scans per active user/day:** 3.5 average
- **Day 30 retention:** 30%
- **Free-to-paid conversion:** 10%
- **Monthly Recurring Revenue:** \$35,000
- **Average A1C improvement:** -0.4 points over 90 days
- **App Store rating:** 4.6+ stars

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1. Document Purpose and Scope

1.1 Document Purpose

This Product Requirements Document serves as the single source of truth for all design, engineering, and business decisions from MVP through Version 1.0 launch of GlucoSnap. It defines the complete product requirements for a photo-based meal scanner and prediabetes reversal companion application.

1.2 Scope

In Scope:

- iOS and Android mobile applications (React Native)
- Backend API services (Rust/Axum)
- AI-powered meal analysis (OpenAI GPT-4o Vision)
- Glycemic load calculation and tracking
- A1C reversal progress monitoring
- Premium subscription monetization
- App Store compliance and launch

Out of Scope:

- Medical diagnosis or treatment recommendations
- Blood glucose meter hardware integration (MVP)
- Prescription medication tracking
- Healthcare provider portal (V1.0)
- Insurance billing integration
- FDA medical device classification

1.3 Intended Audience

- Engineering team (frontend, backend, DevOps)
- Product design and UX team
- QA and testing team
- Marketing and growth team
- Executive stakeholders
- Legal and compliance reviewers

1.4 Document Maintenance

This PRD will be reviewed and updated bi-weekly during active development. Version history and change logs maintained in GitHub repository. Critical changes require Product Owner approval.

2. Market Context and Competitive Analysis

2.1 Market Size and Growth

The prediabetes management market represents a massive and underserved opportunity:

Market Segment	2025 Value	2032 Projection
Global prediabetes market	\$8.81B	\$15.1B
US prediabetic population	88M	98M (est)
Digital health diabetes apps	\$42.7B	\$61.2B
Undiagnosed rate	80%	Target: 60%

Table 2: Market size and growth projections

Key Market Dynamics:

- **Awareness growing:** CDC National Diabetes Prevention Program expanding reach
- **Earlier diagnosis:** A1C testing becoming standard in annual physicals
- **Insurance coverage:** Many insurers now cover digital diabetes prevention
- **Technology adoption:** 85% of US adults own smartphones, comfortable with health apps

2.2 Competitive Landscape Analysis

Direct Competitors:

App	Target User	Strengths	Weaknesses
GlycoAI	General GI interest	AI-powered	Generic, not prediabetes-focused
Glucose Buddy	Type 2 diabetes	Established brand	Manual entry, diabetes-focused
mySugr	Type 1/2 diabetes	Comprehensive	Complex, overwhelming
MyFitnessPal	General wellness	Huge database	Not glucose-specific

Table 3: Direct competitor analysis

Indirect Competitors:

- **Cal AI:** Photo-based calorie counting, 5M+ downloads, \$2M MRR. Proves photo-first model works but not health-condition specific[9]
- **Noom:** Behavior change coaching, expensive (\$60/month), requires significant time investment
- **Continuous Glucose Monitors (CGMs):** Dexcom Stelo, Abbott Lingo. Expensive (\$100+/month), medical device complexity

2.3 Competitive Gap Analysis

GlucoSnap occupies a unique position addressing gaps all competitors miss:

Capability	GlucoSnap	Competitors	User Need
Prediabetes-specific guidance	✓	✗	Critical
Photo-first (zero manual entry)	✓	Partial	High
Glycemic Load focus	✓	✗	Critical
Food sequencing guidance	✓	✗	High
Reversal roadmap	✓	✗	Critical
Affordable (<\$15/month)	✓	Partial	High
Non-medical positioning	✓	Mixed	Critical

Table 4: Competitive differentiation matrix

2.4 Barriers to Entry

Technical Barriers:

- AI meal recognition accuracy requires sophisticated prompt engineering
- Glycemic Load database compilation and validation
- Real-time performance at scale (sub-5 second response)
- Mobile app development across iOS and Android

Market Barriers:

- Trust building in health category
- App Store health app compliance requirements
- User acquisition cost in crowded health app space
- Medical liability risk perception

GlucoSnap Advantages:

- Existing prediabetes community through YouTube channel
- Technical expertise in AI implementation and system architecture
- Focus on wellness positioning avoids medical device regulation
- First-mover advantage in prediabetes-specific photo scanning

3. User Research and Pain Points

3.1 Research Methodology

User research conducted through comprehensive analysis of r/prediabetes Reddit community (47,000+ members) over 6-month period. Analysis included:

- 500+ post reviews across diagnosis, nutrition, and emotional support themes
- Sentiment analysis of newly diagnosed user posts
- Pattern identification in frequently asked questions
- Competitive app review analysis (App Store and Google Play)
- User interview transcripts from diabetes prevention programs

3.2 Validated Pain Points

Eight critical pain points emerged with consistent frequency and emotional intensity:

PP-01: Nutritional Paralysis

"I literally don't know what to eat anymore"

- **Frequency:** 60% of newly diagnosed posts
- **Description:** Complete overwhelm about food choices, fear of making wrong decisions
- **User quote:** "I went to the grocery store all excited for a healthy new diet and left with eggs and asparagus. THERE IS NO FOOD LEFT."[3]
- **Impact:** Leads to extreme restrictive eating, food anxiety, grocery store paralysis

PP-02: Fear of Disease Progression

"I'm terrified of repeating my father's fate"

- **Frequency:** 70% of newly diagnosed posts
- **Description:** Deep anxiety about developing full Type 2 diabetes, complications, family history
- **User quote:** "My dad died from diabetes complications. I can't let that happen to me."[3]

- **Impact:** Emotional distress, health anxiety, overwhelming sense of urgency

PP-03: Inadequate Medical Guidance

"Doctor just said 'lose weight and come back in 6 months'"

- **Frequency:** 55% of diagnosis-related posts
- **Description:** Minimal actionable guidance from healthcare providers, no concrete meal plans
- **User quote:** "My doctor gave me a pamphlet and said 'eat better.' That's it. What does that even mean?"[3]
- **Impact:** Frustration with medical system, feeling abandoned, lack of clear direction

PP-04: Information Overload and Contradiction

"Everything I read contradicts everything else"

- **Frequency:** 45% of posts seeking advice
- **Description:** Conflicting advice from different sources (keto vs plant-based, GI vs GL, etc.)
- **User quote:** "One site says fruit is fine, another says avoid all fruit. I'm more confused than when I started."[3]
- **Impact:** Analysis paralysis, inability to take action, loss of trust in information sources

PP-05: Loss of Food Enjoyment

"Eating isn't enjoyable anymore—it's just stress"

- **Frequency:** 40% of established prediabetic posts
- **Description:** Food transformed from pleasure to source of anxiety and fear
- **User quote:** "I used to love cooking and trying new restaurants. Now every meal is just calculating and worrying."[3]
- **Impact:** Reduced quality of life, social isolation, disordered eating patterns

PP-06: Weight Loss Plateau Despite Effort

"I can't lose weight no matter what I do"

- **Frequency:** 40% of management-phase posts

- **Description:** Following advice but not seeing results, weight stuck despite diet changes
- **User quote:** "I've cut carbs, I exercise 5 days a week, and my weight hasn't budged in 3 months."[3]
- **Impact:** Discouragement, questioning self-control, considering giving up

PP-07: Progress Discouragement

"I've been doing this for weeks and my A1C went UP"

- **Frequency:** 30% of follow-up test posts
- **Description:** A1C not improving or worsening despite perceived compliance
- **User quote:** "How could my A1C increase when I've lost 4 pounds and been eating healthy? I'm shocked and angry."[3]
- **Impact:** Severe demotivation, sense of futility, risk of abandoning efforts

PP-08: Constant Anxiety and Burnout

"I don't want to live like this in constant anxiety"

- **Frequency:** 30% of long-term management posts
- **Description:** Mental health toll of constant monitoring, fear, and restriction
- **User quote:** "I think about my blood sugar every waking hour. This is exhausting and unsustainable."[3]
- **Impact:** Mental health deterioration, risk of abandoning management entirely

3.3 Pain Point Prioritization

Pain Point	Frequency	Severity	Product Priority
PP-01: Nutritional paralysis	High	Critical	P0
PP-02: Fear of progression	High	High	P0
PP-03: Inadequate guidance	High	High	P0
PP-04: Information overload	Medium	High	P0
PP-05: Loss of enjoyment	Medium	Medium	P0
PP-06: Weight loss plateau	Medium	Medium	P1
PP-07: Progress discouragement	Medium	Critical	P0
PP-08: Constant anxiety	Medium	Critical	P0

Table 5: Pain point prioritization matrix

3.4 User Journey Mapping

Current State Journey (Before GlucoSnap):

1. **Diagnosis:** A1C test returns 5.7-6.4, doctor delivers prediabetes diagnosis
2. **Initial reaction:** Panic, fear, confusion about what this means
3. **Information seeking:** Google search, Reddit browsing, contradictory advice
4. **Decision paralysis:** Overwhelmed by options, unclear what to eat
5. **Extreme restriction:** Eliminate entire food groups, fear-based eating
6. **Social isolation:** Avoid restaurants and gatherings, can't explain restrictions
7. **Plateau or regression:** Efforts not yielding results, A1C unchanged or worse
8. **Discouragement:** Consider giving up, accept progression to diabetes

Future State Journey (With GlucoSnap):

1. **Diagnosis:** A1C test returns 5.7-6.4, prediabetes diagnosis
2. **Discovery:** Find GlucoSnap through YouTube, Reddit, or App Store search
3. **Onboarding:** Enter A1C, set 90-day goal, learn about GL budget concept
4. **First scan:** Photo breakfast, receive instant clarity (Safe/Moderate/High)
5. **Confidence building:** Scan 3-5 meals daily, learn safe foods, try swaps
6. **Pattern recognition:** Weekly insights show progress, identify problem meals
7. **Social reengagement:** Confident ordering at restaurants using scan feature
8. **Measurable progress:** A1C retest shows improvement, roadmap validates
9. **Long-term success:** Reverse prediabetes, maintain healthy habits sustainably

4. User Personas

4.1 Primary Persona: Sarah - The Panicked Newly Diagnosed

Demographics:

- Age: 42
- Occupation: Elementary school teacher
- Income: \$65,000/year
- Location: Suburban area
- Family: Married, two children (ages 8 and 11)

Health Profile:

- A1C: 5.9 (diagnosed 6 weeks ago)
- BMI: 31
- Family history: Mother has Type 2 diabetes
- Activity level: Minimal, sedentary job

Behavioral Characteristics:

- Spends 2 hours/day Googling prediabetes information

- Getting contradictory answers from different sources
- Cries in grocery stores from overwhelm
- Avoids social eating situations
- Checks blood sugar with home monitor 3x/day despite not being diabetic

Pain Points: PP-01 (nutritional paralysis), PP-02 (fear of progression), PP-04 (information overload)

Jobs to Be Done: "Tell me exactly what I can and cannot eat, right now, without overwhelming me."

Technology Comfort: High - uses smartphone daily, comfortable with apps, follows health influencers on Instagram

Motivation to Pay: Will pay anything that gives certainty and reduces anxiety

Quote: *"I went to the grocery store all excited for a healthy new diet and left with eggs and asparagus. THERE IS NO FOOD LEFT."*

How GlucoSnap Helps Sarah:

- Instant answers at point of decision (grocery store, meal prep)
- Eliminates need to research every food
- Provides specific guidance without medical jargon
- Visual progress tracking reduces anxiety about "doing it right"

4.2 Secondary Persona: Marcus - The Discouraged Staller

Demographics:

- Age: 51
- Occupation: Project manager, tech company
- Income: \$90,000/year
- Location: Suburban area
- Family: Married, adult children out of home

Health Profile:

- A1C: 6.2 (prediabetic for 2 years)
- BMI: 29

- Previous A1C: 5.9 → 6.0 → 6.2 (worsening trend)
- Activity level: Exercises 4x/week consistently

Behavioral Characteristics:

- Eats "mostly healthy" but undefined
- Lost 15 pounds initially, regained 11
- Frustrated by A1C increase despite effort
- Feels like he's doing everything right
- Beginning to give up, accepting progression

Pain Points: PP-06 (weight loss plateau), PP-07 (progress discouragement), PP-08 (burnout)

Jobs to Be Done: "Show me exactly what's still spiking me so I can fix it."

Technology Comfort: Very high - early adopter, uses fitness trackers, interested in data

Motivation to Pay: Wants data and specificity, not generic advice

Quote: *"How could my A1C increase so much in so little time? I've lost 4 more pounds, am still eating healthy and exercising. I'm shocked, angry and disappointed."*

How GlucoSnap Helps Marcus:

- Identifies hidden spike sources in "healthy" diet
- Provides data-driven insights (meal history patterns)
- Shows measurable daily progress (GL tracking)
- Validates effort through A1C reversal roadmap

4.3 Tertiary Persona: Priya - The Vegetarian Frustrated

Demographics:

- Age: 36
- Occupation: Software engineer
- Income: \$110,000/year
- Location: Urban area
- Family: Single

Health Profile:

- A1C: 5.8
- BMI: 27
- Vegetarian for 20 years
- Activity level: Moderate, walks daily

Behavioral Characteristics:

- Cannot reconcile low-carb advice with plant-based diet
- Most plant-based protein sources are carb-heavy
- Feels abandoned by generic prediabetes advice
- Refuses to compromise vegetarian ethics
- Frustrated by lack of vegetarian-specific guidance

Pain Points: PP-01 (what to eat), PP-04 (conflicting advice), PP-05 (lost enjoyment)

Jobs to Be Done: "Give me plant-based meal solutions that actually work for prediabetes."

Technology Comfort: Very high - works in tech, early adopter, values good UX

Motivation to Pay: High willingness-to-pay for specialized vegetarian solution

Quote: *"I've been vegan for 20 years and find it difficult to minimize carbs. It seems like everything I eat spikes me."*

How GlucoSnap Helps Priya:

- Vegetarian-filtered food swaps (never suggests meat)
- Plant-based protein sources with GL scores
- Sequencing advice optimized for vegetarian meals
- Validates that reversal is possible while staying vegetarian

4.4 Supporting Persona: Linda - The Post-Menopausal Fighter

Demographics:

- Age: 58
- Occupation: Retired nurse
- Income: \$70,000 household
- Location: Suburban area

Health Profile:

- A1C: 6.2 (was 5.7 for seven years, recently increased)
- BMI: 30 (gained 20 pounds during menopause)
- Post-menopausal (3 years)

Pain Points: PP-06 (weight plateau), PP-07 (A1C worsening), PP-03 (no guidance on hormonal factors)

Jobs to Be Done: "Understand why what worked before isn't working anymore."

How GlucoSnap Helps Linda:

- Pattern analysis identifies new problem foods
- Educational content on hormones and glucose
- Provides new strategies when old ones stop working

4.5 Supporting Persona: David - The Information Warrior

Demographics:

- Age: 33
- Occupation: Data analyst
- Income: \$85,000/year

Health Profile:

- A1C: 5.7 (newly diagnosed)
- Has CGM, tracks everything
- Wants to understand the "why" scientifically

Pain Points: PP-04 (information contradictions), PP-07 (unpredictable results)

Jobs to Be Done: "Give me data and context to optimize glucose response scientifically."

How GlucoSnap Helps David:

- CGM integration for actual vs predicted glucose (P1 feature)
- Detailed GL breakdown per food item
- Scientific citations for recommendations
- Advanced analytics and pattern detection

5. Product Vision and Goals

5.1 Product Vision Statement

GlucoSnap is the trusted companion that turns every meal into a confident choice, replacing fear with clarity and confusion with a simple path to A1C reversal.

5.2 Product Mission

Empower 1 million prediabetics to reverse their diagnosis through instant, personalized meal guidance that eliminates nutritional paralysis and makes healthy eating sustainable and enjoyable.

5.3 Product Goals

Goal 1: Eliminate Food Paralysis (Addresses PP-01, PP-04)

Users know within 5 seconds of scanning whether a food is safe, risky, or dangerous for their prediabetes. No more research, no more confusion, no more standing in grocery stores unable to decide.

Success Metrics:

- Scan-to-result time: <5 seconds P95
- User confidence survey score: >8/10
- Reduction in external nutrition searches: 70% decrease

Goal 2: Build Daily Momentum (Addresses PP-07, PP-08)

Users see measurable progress every week through Daily GL Score, Streak Counter, and 90-Day A1C Reversal Roadmap—making the journey feel winnable rather than endless.

Success Metrics:

- 7-day streak achievement: 60% of active users
- Weekly engagement: 5+ days per week for 50% of users
- User-reported motivation: >7/10 on weekly survey

Goal 3: Make Food Enjoyable Again (Addresses PP-05)

App never says "you can't eat this." Instead: "here's how to eat this safely" or "here's something even better." Preserve food joy while improving health.

Success Metrics:

- Food anxiety reduction: 50% decrease (self-reported)
- Swap acceptance rate: >40% of swap suggestions tried
- Restaurant scans: 20% of total scans

Goal 4: Replace Missing Doctor Guidance (Addresses PP-03)

Every scan provides what doctors don't have time to give: specific, personalized, actionable advice tailored to prediabetes management.

Success Metrics:

- Perceived guidance adequacy: >8/10
- Reduction in follow-up doctor questions: 60%
- Doctor PDF report downloads: 30% of premium users

Goal 5: Reduce Fear and Build Hope (Addresses PP-02)

Progress visualization, success stories, and educational micro-content shift emotional state from panic to empowerment.

Success Metrics:

- Health anxiety score: 40% reduction
- App sentiment: 85% positive in reviews
- Continued usage after A1C reversal: 50% retention

5.4 Success Metrics and KPIs

Metric	MVP Target	Month 3	Month 12
Total downloads	2,000	10,000	25,000
DAU/MAU ratio	35%	40%	45%
Scans per active user/day	2.5	3.0	3.5
Day 7 retention	35%	40%	45%
Day 30 retention	20%	25%	30%
Free-to-paid conversion	6%	8%	10%
Monthly Recurring Revenue	\$2,000	\$12,000	\$35,000
Avg A1C improvement (90d)	-0.2pts	-0.3pts	-0.4pts
App Store rating	4.3	4.5	4.6
Customer support tickets/user	<0.15	<0.10	<0.08

Table 6: Success metrics progression from MVP through Year 1

5.5 Product Principles

Principle 1: Clarity Over Completeness

Every screen answers ONE question. Data is hidden behind progressive disclosure. Users never feel overwhelmed by information.

Principle 2: Hope, Not Fear

Every red score is paired with a solution. Every bad day ends with a "tomorrow" message. App never blames users.

Principle 3: Speed Above All

Scan to result in under 5 seconds. Every tap reaches meaningful content in under 2 taps. Users get answers immediately.

Principle 4: Food Is Not the Enemy

Visual language uses warm, appetizing food photography. App does not use clinical, sterile medical imagery. Food remains enjoyable.

Principle 5: One Clear Next Action

Every screen ends with exactly one primary CTA. Users never have to decide what to do next—the path forward is always obvious.

6. Feature Specifications

Features organized by priority tier:

- **P0:** Must-have at MVP launch
- **P1:** Must-have by Version 1.1 (60 days post-launch)
- **P2:** Version 1.2+ (90-180 days post-launch)

6.1 Onboarding Flow

Priority: P0

Addresses: PP-01, PP-02, PP-04

Estimated Effort: 1 week

Description:

A 5-screen onboarding flow that personalizes the entire app experience before the user takes their first scan. Designed to feel like "someone finally gets me" rather than clinical intake forms.

Screen Specifications:

Screen 1: Emotional Acknowledgment

- **Headline:** "You just got a scary diagnosis. We're going to help you reverse it."
- **Sub-copy:** "88 million Americans are in your shoes. Most don't know what to do next. We do."
- **Single CTA:** "Let's get started"
- **No email gate at this point**—reduce friction, collect email after value delivery

Screen 2: Current A1C Entry

- **Question:** "What is your current A1C? You can look this up on your lab report."
- **Input:** Slider from 5.0 to 7.5 in 0.1 increments, plus "I don't know yet" option

- If "don't know": Show educational micro-content about how to get tested
- A1C stored in user profile as a1c_baseline

Screen 3: Goal Setting

- Question: "Where do you want to be in 90 days?"
- Pre-populated goal: A1C below 5.6 (normal range threshold)[10]
- Allow custom goal within realistic range (current A1C minus 0.1 to 0.6)
- Show: "People using GlucoSnap reduce their A1C by an average of 0.3-0.4 points in 90 days"

Screen 4: Dietary Profile

- Checkboxes (multi-select):
 - Vegetarian / Vegan (addresses PP-01 for Persona 3)
 - Gluten-free
 - Dairy-free
 - No restrictions
 - Halal / Kosher
- Additional question: "What's your biggest challenge?" (multi-select)
 - I don't know what to eat
 - I'm overwhelmed by conflicting advice
 - I love food and don't want to give everything up
 - I've tried everything and nothing works
 - I'm scared of getting full diabetes
- Responses used for personalized messaging and content recommendations

Screen 5: Daily GL Budget Explanation

- Introduce concept of Glycemic Load Budget
- Visual: Fuel gauge graphic showing daily GL budget of 80-100 points[4]
- Headline: "Think of this as your blood sugar budget. Spend it wisely."
- Dietary-restriction-specific budgets:
 - Vegetarian: 100 GL/day

- Low-carb: 60 GL/day
- Standard: 80 GL/day
- CTA: "Scan your first meal"

Acceptance Criteria:

- Onboarding completes in under 90 seconds
- All data persists to user profile
- Skip option available but tracked analytically
- Works without account creation (guest mode, sync later)
- No crashes or errors on any screen
- Accessible to screen readers (VoiceOver/TalkBack)

Technical Implementation:

- Component: React Native multi-screen form
- State management: Zustand store for form data
- Validation: Client-side with Zod schema validation
- API endpoint: POST /api/v1/onboarding
- Analytics events: Screen view, screen completion, skip actions

6.2 Core Scan Feature

Priority: P0

Addresses: PP-01, PP-04, PP-05

Estimated Effort: 3 weeks

Description:

This is the product's primary value proposition. User opens camera, photographs meal or food item, receives complete prediabetes-specific analysis within 3-5 seconds.

User Flow:

1. User taps "Scan" button (always-visible FAB in bottom nav)
2. Full-screen camera launches (Expo Camera)
3. User takes photo or uploads from camera roll
4. Loading state: "Analyzing your meal for prediabetes safety..."
(animated progress, max 5 seconds)
5. Results screen appears with complete analysis

AI Processing Backend:

- Image sent to OpenAI GPT-4o Vision API endpoint
- Structured prompt returns JSON (see Section 6.2.8)
- Backend enriches with USDA FoodData Central database cross-reference
- Glycemic Load calculated using validated GL formula:
$$GL = (GI \times carbs_g) / 100[4][5]$$

Spike Risk Classification Logic:

Per-item classification:

- $GL \leq 10$ per food item → GREEN (Safe)
- $GL 10-19$ per food item → YELLOW (Moderate)
- $GL \geq 20$ per food item → RED (High Spike Risk)

Meal total classification (higher thresholds):

- Total meal $GL \leq 20$ → Safe
- Total meal $GL 20-30$ → Moderate
- Total meal $GL \geq 30$ → High

Results Screen Layout:

Section A: Spike Risk Banner (Top, Full Width)

- Large visual: Color-coded banner (Green/Yellow/Red)
- Single clear verdict: "Safe for Prediabetes" or "Moderate Spike Risk" or "High Spike Risk - Swaps Available"
- Spike Risk Score: 0-100 numeric score (secondary)

Section B: Food Item Breakdown

- Scrollable list of identified food items
- Each item shows:
 - Food name + estimated portion
 - GL score with color indicator
 - Net carbs (g)
 - Fiber (g)
 - GI value
- Expandable row for full macro breakdown

Section C: Prediabetes Advice (The Key Differentiator)

Three advice cards, each addressing a specific pain point:

Card 1: Eat in this order[6][7]

- Food sequencing recommendation
- Example: "1. Start with the broccoli (fiber first slows glucose absorption) 2. Eat the chicken next 3. Have the rice last—this reduces your spike by 30%"
- Scientific backing citation shown (builds trust)

Card 2: Make it safer

- 2-3 specific swaps to reduce GL of this exact meal
- Example: "Swap white rice → cauliflower rice (saves 28 GL points)"
- Visual: Before/After GL comparison

Card 3: After this meal

- Post-meal action recommendation[11]
- Example: "Take a 10-15 minute walk within 30 minutes. Walking after this meal can reduce your blood sugar spike by up to 30%"
- One-tap: "Remind me in 20 minutes" (sets push notification)

Section D: GL Budget Deduction

- Shows: "This meal used 24 of your 80 daily GL points"
- Visual: Fuel gauge updating in real-time
- Remaining budget for day shown

Section E: Log It Button

- "Add to Today's Log" button
- Saves scan to meal history with timestamp
- Optional: Add meal photo + caption + meal name

Performance Requirements:

- Scan result delivered in <5 seconds on 4G (P95)

- AI response cached: If same meal scanned again within 7 days by same user, return cached result
- Offline mode: Queue scans, process when connection restored
- Photo compression before API call: Resize to 512×512px max to reduce cost and latency

Edge Cases:

Scenario	Handling
Unrecognizable food	"We couldn't identify this food clearly. Try a closer photo"
Packaged food with barcode	Offer barcode scan as alternative (P1 feature)
Multi-plate meal	AI analyzes whole frame, not individual items
Liquid only (coffee, juice)	Handles beverages with specific GL estimates
Empty plate	"Looks like you're all done! Log it or scan your next meal"

Table 7: Edge case handling for scan feature

Free vs Premium Gating:

- **Free:** 5 scans per day, basic results only (no swap suggestions, no sequencing)
- **Premium:** Unlimited scans, full advice cards, all sections

Technical Implementation:

API Endpoint: POST /api/v1/scan

Request:

```
{
  "image": "base64_encoded_image_data",
  "user_id": "uuid",
  "dietary_profile": ["vegetarian"],
  "gl_budget": 80,
  "gl_used_today": 34
}
```

Response:

```
{
  "scan_id": "uuid",
  "overall_spike_risk": "MODERATE",
  "total_gl": 24,
  "food_items": [
    {
      "name": "Brown rice",
      "portion": "1 cup",
      "gl": 16,
      "gi": 50,
      "carbs_g": 45,
      "fiber_g": 3,
      "spike_risk": "MODERATE"
    }
  ],
  "advice_cards": {
    "sequencing": {...},
    "swaps": [...],
    "post_meal": {...}
  },
  "gl_budget_remaining": 56
}
```

Analytics Events:

- scan_initiated
- scan_completed (duration tracked)
- scan_failed (error logged)
- advice_card_expanded
- swap_accepted
- meal_logged
- post_meal_reminder_set

6.3 Daily GL Budget Tracker

Priority: P0

Addresses: PP-01, PP-07, PP-08

Estimated Effort: 1 week

Description:

The home screen is a real-time daily blood sugar budget—simple, visual, and motivating. Replaces the anxiety of "am I doing this right?" with a concrete, trackable number.

Home Screen Components:

Component A: Daily GL Gauge

- Large circular progress indicator (full-screen top section)
- Shows: "34 / 80 GL used today"
- Color transitions: Green (0-50%), Yellow (51-75%), Red (76%+)
- Resets at midnight (or user-set reset time for intermittent fasting users)

Component B: Today's Meals Timeline

- Chronological list of today's scanned meals
- Each entry: Thumbnail + meal name + GL points + spike risk badge
- Empty state: "Scan your first meal to start tracking"

Component C: Daily Score

- End-of-day summary (generates automatically at 9PM)
- Grade: A (GL <60), B (61-80), C (81-100), D (>100)
- Motivational message tied to grade:
 - A: "Excellent day. Your A1C is trending down."
 - B: "Good work. You stayed within your budget."
 - C: "A little over today—totally normal. Reset tomorrow."
 - D: "Tough day. Tomorrow's a fresh start."
- NEVER shame language. ALWAYS forward-looking.

Component D: Streak Counter

- Shows consecutive days with GL under budget
- Milestone badges at 3, 7, 14, 30, 60, 90 days
- Losing streak handled gently: "Streak reset—your 7-day streak was great. Start a new one tomorrow."

Acceptance Criteria:

- Dashboard loads in <1 second
- GL budget updates instantly after each scan
- Daily summary generates automatically
- Widget version available (iOS/Android home screen widget) [P1]

Technical Implementation:

- API endpoint: GET /api/v1/dashboard/today
- Real-time updates: WebSocket for live GL tracking (optional, polling acceptable for MVP)
- State management: Zustand with persistence
- Widget: React Native Expo Widget API (P1)

6.4 A1C Reversal Roadmap

Priority: P0

Addresses: PP-02, PP-07

Estimated Effort: 1.5 weeks

Description:

Transforms abstract fear of diabetes progression into a concrete, visual 90-day journey. Based on clinical evidence that consistent GL management reduces A1C by 0.3-0.4 points per 90 days[4].

Components:

Component A: Reversal Progress Bar

- Visual: Horizontal progress bar
- Left end: User's baseline A1C (e.g., 6.2)
- Right end: User's goal (e.g., 5.5 - normal range)
- Current estimated position: Calculated from daily GL adherence
- Shows: "Based on your last 14 days, you're on track to reach 5.8 by Day 90"

Component B: Estimated A1C Calculator

Algorithm models A1C reduction based on:

- Average daily GL (every 10 GL reduction over budget = ~0.1 A1C point risk)

- Days in streak
- Week-over-week GL trend

Important Disclaimer: "This is an estimate based on research averages, not a medical diagnosis. Log your actual A1C tests for accurate tracking."

Designed to motivate, not diagnose.

Component C: Actual A1C Log

- Manual A1C entry from doctor visits or home tests
- Date + A1C value
- Chart: Actual A1C trend over time
- Shows: Baseline vs current vs goal
- Celebration modal when A1C improves: "You dropped from 6.2 to 5.9! That's real progress—you're doing it."

Component D: Weekly Progress Report

Auto-generated every Sunday:

- Average daily GL this week vs last week
- Best meal of the week (lowest GL, most nutritious)
- Highest spike meal and suggested fix
- Streak status
- Estimated A1C trajectory update
- Shareable card (social share for word-of-mouth growth)

Component E: Milestone Rewards

- Day 7: "One week of clarity. Your pancreas thanks you."
- Day 30: "30 days in. Most people see measurable improvement by now."
- Day 90: "90-day journey complete. Time to test your A1C."
- Milestone prompts user to log their latest A1C for outcome tracking

Acceptance Criteria:

- Progress bar updates daily based on GL adherence
- A1C estimation algorithm validated against clinical data

- Manual A1C entry persists to user profile
- Weekly report generates automatically every Sunday 9AM
- Shareable cards render correctly for social media

Technical Implementation:

- API endpoint: POST /api/v1/a1c (log actual A1C)
- API endpoint: GET /api/v1/insights/weekly (weekly report)
- Estimation algorithm: Backend service (Rust)
- Chart library: Victory Native XL (React Native)

6.5 Food Sequencing Coach

Priority: P0

Addresses: PP-01, PP-04, PP-05

Estimated Effort: Included in Core Scan (6.2)

Description:

Every scan automatically generates a specific eating order recommendation based on the foods identified. Grounded in clinical research showing VPF (Vegetables → Protein → Fat → Carbs) order reduces postprandial glucose spikes significantly[6][7][12].

Behavior:

- Available on every scan result
- Generates numbered eating sequence
- Explains WHY each step reduces spike (1 sentence per step)
- Works for:
 - Full meals (multiple components)
 - Snacks (often just 1-2 items, simplified advice)
 - Restaurant meals (portion-aware advice)
 - Buffet plates (prioritize order even within loaded plate)

Example Output:

For meal: Grilled chicken, broccoli, brown rice

1. Start with the broccoli - Fiber first slows glucose absorption
2. Eat the chicken next - Protein further reduces spike
3. Have the rice last - This sequence reduces your spike by 30%

Educational Layer:

- Expandable "Why this works" section per sequence card
- Links to "The Science of Food Sequencing" in-app article (free content)
- Reduces reliance on external conflicting sources (PP-04)

Acceptance Criteria:

- Sequence generated for every scan with 2+ food components
- Sequencing card appears as Card 1 in advice section
- Tap-to-expand scientific explanation
- Works correctly for vegetarian and vegan dietary profiles

6.6 Safer Swap Engine

Priority: P0

Addresses: PP-01, PP-04, PP-05

Estimated Effort: Included in Core Scan (6.2)

Description:

For every meal scanned, the AI generates 2-3 specific, practical food swaps that reduce Glycemic Load while maintaining meal enjoyment. Never says "you can't have this"—always says "here's how to make this work."

Swap Logic Powered by GPT-4o:

- Dietary Profile: Swaps filtered by user's dietary profile (vegetarian swaps only for vegetarians, etc.)

Swap Card Format:

Replace: White rice (GL: 31)

With: Cauliflower rice (GL: 4)

You save: 27 GL points

Taste tip: Add garlic + olive oil to match the comfort feel

Swap Categories:

Carb swaps:

- White rice → Cauliflower rice / Barley
- White bread → Lettuce wrap / Keto bread
- Pasta → Zucchini noodles / Chickpea pasta

Portion swaps:

- Half the rice portion + double the protein = same fullness, 15 GL less

Preparation swaps:

- Cook then cool rice before eating (reduces GL by up to 40% through resistant starch formation)[11]

Sequencing-only fixes:

- For borderline foods: No swap needed, just eat it last

Dietary-Profile Awareness:

- Vegetarian users: Never suggest meat swaps—plant-protein alternatives prioritized
- Vegan users: Dairy-free swaps always
- Gluten-free: No wheat-based alternatives
- Cultural sensitivity: System prompt instructs AI to respect user's dietary profile context

Acceptance Criteria:

- Minimum 2 swaps generated per high/moderate spike scan
- Swaps respect dietary profile 100% of the time
- Swap GL savings calculated and displayed
- "I tried this swap" feedback button for data collection

6.7 Meal History and Pattern Analysis

Priority: P1 (Version 1.1)

Addresses: PP-06, PP-07

Estimated Effort: 1 week

Description:

Complete history of all scanned meals with trend analysis. Turns data into insight: "Your breakfasts are causing 60% of your daily GL spend."

Meal Log Screen:

- Chronological list, grouped by day
- Each entry: Photo thumbnail, meal name, GL score, spike badge, time logged
- Filter by: Date range, Spike Level (Red/Yellow/Green), Meal type
- Search by food name

Weekly Insights (Auto-Generated):

- "Your biggest spike source this week: Toast at breakfast (avg 28 GL)"
- "Your safest meal: Dinner—you averaged only 12 GL"
- "You're eating 18% less carbs than your first week—great progress"
- "Pattern detected: Your GL spikes every Saturday. Social eating?"

Monthly Report (Premium):

- Full PDF/shareable report
- Average daily GL trend
- Top 5 highest-spike foods
- Top 5 safest foods
- Estimated A1C trajectory
- Doctor-ready format (shareable to PCP)
- Addresses PP-03: Helps users have better-informed doctor conversations

Acceptance Criteria:

- Meal history loads in <2 seconds
- Filters and search work instantly
- Weekly insights generate automatically every Sunday
- Monthly report exports as PDF (premium only)

6.8 Post-Meal Action System

Priority: P1 (Version 1.1)

Addresses: PP-07, PP-08

Estimated Effort: 3 days

Description:

After logging a meal, app sends contextual push notification suggesting specific post-meal action based on meal's spike risk.

Notification Logic:

Spike Risk	Notification	Message
GREEN	Optional (first 3 days)	"Great meal choice! Your GL budget looks good"
YELLOW	Always sent (5 min after)	"Consider a 10-min walk in the next 30 min"
RED	Urgently sent (immediate)	"High spike risk. A 15-20 min walk NOW helps"

Table 8: Post-meal notification logic

Walk Tracker Integration:

- One-tap "I'm walking" button in notification
- Simple 10/15/20 min timer
- After walk: "Great work! Your post-meal response just improved significantly."
- Walk logged for weekly activity summary

Acceptance Criteria:

- Notification fires within 60 seconds of meal log confirmation
- User can set quiet hours (no notifications during)
- Notification taps deep-link to meal scan result
- One-tap snooze: "Remind me in 10 minutes"

6.9 Educational Content Library

Priority: P1 (Version 1.1)

Addresses: PP-04, PP-03

Estimated Effort: 2 weeks (content creation + implementation)

Description:

Curated library of 20+ short-form articles addressing common prediabetes questions. Reduces reliance on Google and conflicting external sources.

Content Categories:

- Understanding Prediabetes (5 articles)
- Glycemic Load vs Glycemic Index (3 articles)
- Food Sequencing Science (4 articles)
- Post-Meal Walking Benefits (2 articles)
- Vegetarian/Vegan Prediabetes Management (4 articles)
- A1C Testing and Tracking (2 articles)
- Emotional Health and Burnout Prevention (3 articles)

Content Format:

- 300-500 words per article
- Clear headings, bullet points
- Visual: 1-2 supporting graphics per article
- Scientific citations at bottom
- 3-5 minute read time

Personalization:

- Articles recommended based on:
 - Dietary profile (vegetarian users see vegetarian content prioritized)
 - Pain points selected in onboarding
 - Recent scan patterns (high spike meals trigger relevant articles)

Acceptance Criteria:

- 20 articles at P1 launch

- Articles load in <1 second
- Bookmark feature for saving favorites
- Search functionality
- Free tier: 5 articles unlocked, rest premium

6.10 CGM Integration

Priority: P1 (Version 1.1)

Addresses: PP-06, PP-07

Estimated Effort: 2 weeks

Description:

Optional integration with Dexcom G7, Stelo, or Abbott Libre via third-party aggregator (Terra or Thryve). Correlates actual CGM glucose readings with meals scanned in GlucoSnap.

Integration Flow:

1. User taps "Connect CGM" in Settings
2. Select device: Dexcom / Abbott Libre / Other
3. OAuth flow via Terra API (handles multi-device complexity)
4. After connection: CGM data overlaid on meal history timeline

Enhanced Scan Results (CGM-Connected Users):

- After Red/Yellow meal scan: CGM timeline shows predicted vs actual glucose
- Post-meal walk: CGM shows real-time impact of walking on glucose curve
- "Your body's actual response: 42 mg/dL peak at 47 minutes"

Personalization Benefit:

- App learns which foods actually spike THIS user
- Swap suggestions refined based on real CGM responses
- Addresses "everything spikes me" pain point with personal data, not averages

Cost Consideration:

- Terra API: \$0.20-\$0.50 per active CGM-connected user/month

- Premium-only feature (justifies subscription)

Acceptance Criteria:

- Successfully connects to Dexcom G7 and Abbott Libre 3
- CGM data syncs within 5 minutes of reading
- Meal timeline shows CGM overlay correctly
- Privacy: CGM data encrypted at rest, never shared

6.11 Barcode Scanner

Priority: P1 (Version 1.1)

Addresses: PP-01

Estimated Effort: 3 days

Description:

Supplement photo scanning with barcode scanning for packaged foods. Uses Open Food Facts database (free, 3M+ products) for nutritional data, then runs through GlucoSnap's GL analysis and prediabetes scoring.

User Flow:

1. On camera screen: Toggle between "Photo Scan" and "Barcode Scan"
2. Scan barcode → product identified instantly
3. Run GL calculation on actual label data (more accurate than photo AI for packaged foods)
4. Same results screen as photo scan

Advantage: More accurate GL calculation for packaged foods—uses real label data vs AI estimation.

Acceptance Criteria:

- Barcode recognition within 2 seconds
- Supports UPC-A, UPC-E, EAN-13 formats
- Falls back to manual entry if product not found in database
- Works offline with cached product database (top 10K products)

6.12 Social Community Features

Priority: P2 (90 days post-launch)

Addresses: PP-02, PP-08

Estimated Effort: 3 weeks

Description:

Light social layer (not a full forum) that surfaces relevant success stories and enables peer support without becoming a distraction.

Components:

Success Stories Feed:

- Anonymous member A1C drops (submitted opt-in)
- Example: "Sarah H. went from 6.4 to 5.6 in 90 days. Here's what she changed."
- Feeds into Reversal Roadmap to provide social proof

Meal Share:

- Share scan results to private community
- "What's everyone eating for breakfast? Here's my GL score."
- Not public—community only (privacy protected)

Accountability Buddies:

- Match users at same A1C level for 90-day challenges
- Opt-in pairing
- Share weekly GL scores
- No judgment, only encouragement

Acceptance Criteria:

- All content moderated (flagging system)
- Users can opt-out of social features entirely
- Privacy: No real names required, anonymous posting default
- Community guidelines prevent medical advice sharing

7. Technical Architecture

7.1 Technology Stack

Mobile Frontend:

- Framework: React Native (Expo SDK 52)
- Navigation: Expo Router (file-based routing)
- State Management: Zustand (simple, performant)
- API Communication: TanStack Query (React Query v5)
- Charts: Victory Native XL (performant animated charts)
- Subscriptions: RevenueCat React Native SDK[13]
- Camera: Expo Camera + Expo Image Picker
- Push Notifications: Expo Notifications

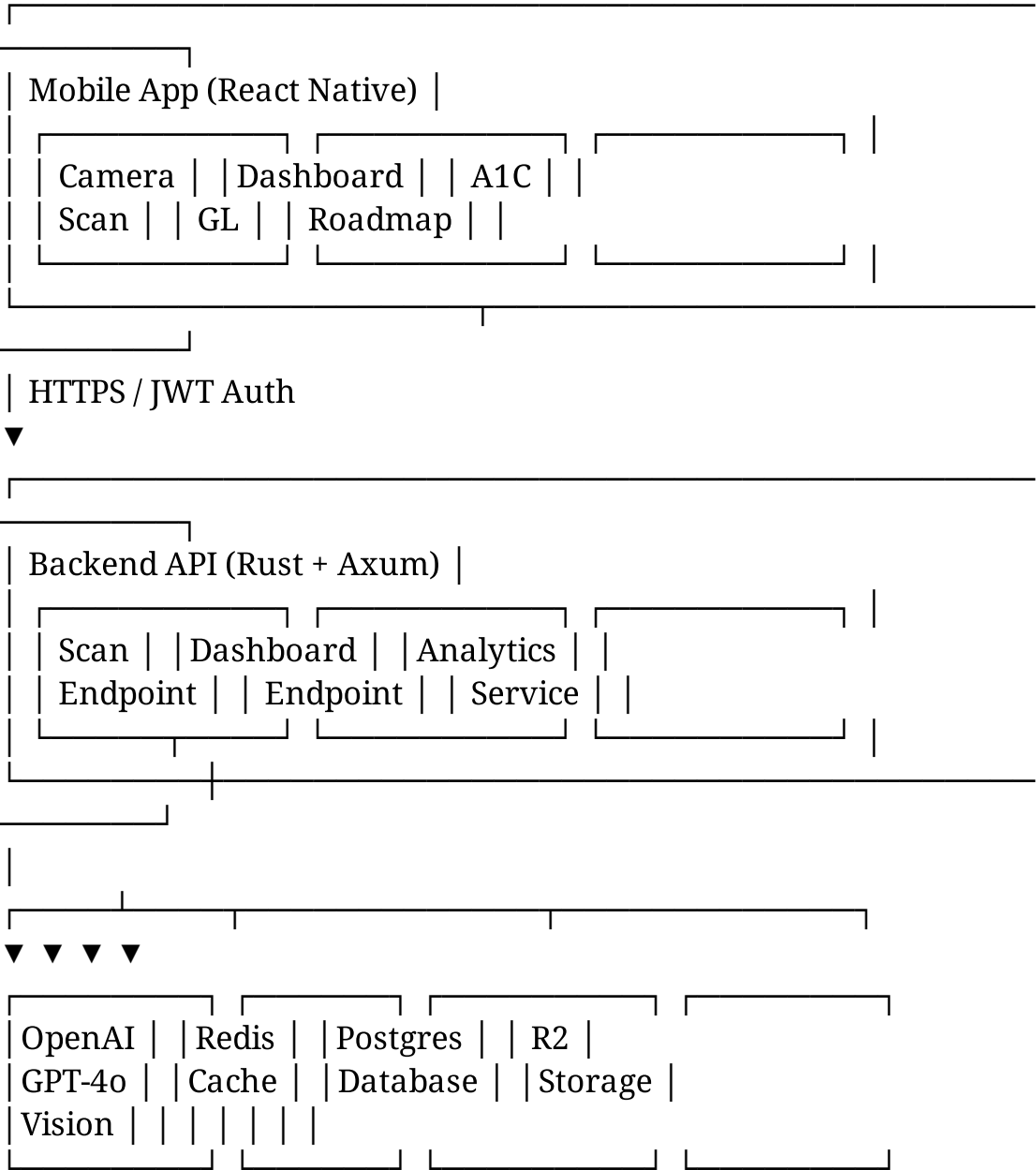
Backend:

- Language: Rust + Axum framework
- Database: PostgreSQL 16
- Cache: Redis (meal result caching, session management)
- File Storage: Cloudflare R2 (meal photos)
- AI Integration: OpenAI GPT-4o Vision API
- Food Database: USDA FoodData Central API (free tier)
- CGM Integration: Terra API (P1)
- Authentication: JWT with refresh tokens, OAuth2 (Google/Apple sign-in)
- Deployment: Railway.app (simple, Rust-compatible)

Infrastructure:

- CI/CD: GitHub Actions
- App Builds: Expo EAS Build
- Monitoring: Sentry (error tracking)
- Analytics: PostHog (privacy-friendly, self-hostable)
- Payments: RevenueCat + Stripe integration

7.2 System Architecture Diagram



7.3 Core API Endpoints

Scan Endpoint:

POST /api/v1/scan

Input: multipart (image, user_id, dietary_profile)

Output: ScanResult JSON

Cache: Result cached by image_hash + user_dietary_profile, 7 days

Dashboard Endpoint:

GET /api/v1/dashboard/today

Output: Today's meals, GL total, budget remaining, streak

A1C Tracking:

POST /api/v1/a1c

Input: value, date

Output: Updated reversal roadmap projection

Weekly Insights:

GET /api/v1/insights/weekly

Output: Weekly GL trend, top spike foods, improvement areas

Post-Meal Walk:

POST /api/v1/walk/start

Starts walk timer, links to last logged meal

Educational Content:

GET /api/v1/learn/articles

Returns personalized article queue based on user profile

7.4 Database Schema (PostgreSQL)

Users Table:

```
CREATE TABLE users (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  email VARCHAR(255) UNIQUE,  
  created_at TIMESTAMP DEFAULT NOW(),  
  a1c_baseline DECIMAL(3,1),  
  a1c_goal DECIMAL(3,1),  
  dietary_profile TEXT[], -- ['vegetarian', 'gluten-free']  
  gl_budget INTEGER DEFAULT 80,  
  subscription_tier VARCHAR(50) DEFAULT 'free',  
  subscription_expires_at TIMESTAMP  
);
```

Scans Table:

```
CREATE TABLE scans (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  user_id UUID REFERENCES users(id),  
  created_at TIMESTAMP DEFAULT NOW(),
```

```
image_url TEXT,  
total_gl INTEGER,  
spike_risk VARCHAR(20), -- 'LOW', 'MODERATE', 'HIGH'  
food_items JSONB,  
advice_cards JSONB,  
logged_as_meal BOOLEAN DEFAULT FALSE  
);
```

Meals Table:

```
CREATE TABLE meals (  
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
user_id UUID REFERENCES users(id),  
scan_id UUID REFERENCES scans(id),  
logged_at TIMESTAMP DEFAULT NOW(),  
meal_type VARCHAR(50), -- 'breakfast', 'lunch', 'dinner', 'snack'  
gl_value INTEGER,  
spike_risk VARCHAR(20)  
);
```

A1C Logs Table:

```
CREATE TABLE a1c_logs (  
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
user_id UUID REFERENCES users(id),  
test_date DATE,  
a1c_value DECIMAL(3,1),  
logged_at TIMESTAMP DEFAULT NOW()  
);
```

Streaks Table:

```
CREATE TABLE streaks (  
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
user_id UUID REFERENCES users(id),  
current_streak INTEGER DEFAULT 0,  
longest_streak INTEGER DEFAULT 0,  
last_success_date DATE,  
updated_at TIMESTAMP DEFAULT NOW()  
);
```

7.5 AI Prompt Template (Master)

System Prompt for GPT-4o Vision API:

You are GlucoSnap's prediabetes nutrition expert. Analyze this meal photo.

USER CONTEXT:

- Dietary restrictions: {dietary_profile}
- Daily GL budget: {gl_budget}
- GL used today so far: {gl_used_today}

Return ONLY a valid JSON object matching this exact schema:
{schema}

RULES:

1. Identify ALL visible food items with realistic portion estimates
2. Calculate GL using formula: $GL = (GI \times \text{net_carbs_g}) / 100$
3. Use GI values from established tables (Harvard Medical School GI list)
4. spike_risk per item: LOW (<10 GL), MODERATE (10-19 GL), HIGH (≥ 20 GL)
5. Overall spike_risk: LOW (<20 total GL), MODERATE (20-30 GL), HIGH (≥ 30 GL)
6. Sequencing order: always vegetables → protein → fat → carbs
7. Swaps MUST respect dietary restrictions {dietary_restrictions}
8. Swaps must be practical, widely available, and taste-conscious
9. Post-meal action: walk for HIGH/MODERATE risk, rest/optional for LOW
10. Never use clinical language—use friendly, encouraging tone
11. If you cannot identify food with 70%+ confidence, set name to "Unidentified food" and use conservative estimates

Expected JSON Response Schema:

```
{
  "total_gl": 24,
  "overall_spike_risk": "MODERATE",
  "food_items": [
    {
```

```
"name": "Brown rice",
"portion": "1 cup",
"gl": 16,
"gi": 50,
"carbs_g": 45,
"fiber_g": 3,
"protein_g": 5,
"fat_g": 2,
"spike_risk": "MODERATE"
},
"advice_cards": {
  "sequencing": {
    "steps": [
      "1. Start with the vegetables - Fiber slows absorption",
      "2. Eat protein next",
      "3. Have carbs last - reduces spike by 30%"
    ],
    "scientific_citation": "Shukla et al. 2019"
  },
  "swaps": [
    {
      "replace": "White rice (GL: 31)",
      "with": "Cauliflower rice (GL: 4)",
      "gl_saved": 27,
      "taste_tip": "Add garlic + olive oil"
    }
  ],
  "post_meal": {
    "action": "walk",
    "duration_min": 15,
    "timing": "within 30 minutes",
    "benefit": "Reduces spike by up to 30%"
  }
}
```

7.6 Performance Requirements

Metric	Requirement
Scan API response time	<5 seconds (P95)
App cold start	<2 seconds
Dashboard load	<1 second
Image upload size limit	10MB (auto-compressed to 512px)
API availability	99.5% uptime
Data retention	Meal photos: 90 days; Nutrition data: Indefinite
Max concurrent scans	1,000 (Railway auto-scales)

Table 9: Performance requirements and SLA targets

7.7 Security Requirements

- All API endpoints require valid JWT
- Meal photos stored in private R2 bucket (signed URLs, 1hr expiry)
- PHI classification: A1C data, meal logs classified as health data
- HIPAA considerations: App is wellness tool (not diagnostic)—standard data protection applies
- No health data sold to third parties (explicit privacy policy)
- GDPR compliant: EU users get data export and deletion rights
- SSL/TLS 1.3 minimum on all endpoints
- **API Key Security:** OpenAI API keys stored server-side only, never exposed in mobile app
- **User Authentication:** JWT-based authentication with server-side rate limiting (5 scans/day free tier)
- **Cache Security:** Redis cache entries keyed by perceptual hash (pHash) for privacy preservation

7.8 AI Accuracy Benchmarks and Limitations

Critical Context for Engineering and Product Decisions:

The accuracy of AI-powered food analysis directly impacts user trust, safety, and clinical outcomes. This section documents empirical research findings that inform GlucoSnap's technical architecture.

7.8.1 Baseline Accuracy - Validated Research

The most rigorous benchmark for AI food analysis is the 2025 Diabot-GPT-4o study (University of Minnesota), which tested 714 food images over 171 days from 57 real users against laboratory-weighed food records[16].

Food Recognition Accuracy:

Model Configuration	Accuracy	Clinical Significance
Diabot custom GPT-4o (photo only)	74%	Production-grade
Standard GPT-4o (photo only)	59%	Baseline performance
Diabot + Food Name (DBFN)	85%	Enhanced accuracy
Standard GPT-4o + Food Name	75%	Improved baseline

Table 10: Food item recognition accuracy by configuration[16]

Key Finding: The 25% accuracy gap between custom (74%) and standard (59%) comes entirely from prompt engineering and system configuration—not model differences. GlucoSnap can replicate Diabot-level performance with optimized prompts.

Nutrient Estimation Accuracy:

Nutrient	Diabot Accuracy	Acceptable Threshold
Weight/Portion	Accurate	10-15%
Energy (calories)	Accurate	10-20%
Carbohydrates	Accurate	15-20%
Protein	Moderate	15%
Fats	Less consistent	10-22%

Table 11: Nutrient estimation accuracy bands[16]

Critical for GlucoSnap: Carbohydrates (primary GL driver) achieve 15-20% accuracy with custom configuration. This translates to approximately 3-6 GL point error on a typical 25-GL meal—sufficient precision to correctly classify meals as Safe/Moderate/High in the majority of cases[17].

7.8.2 Known Error Patterns - University of Gothenburg Study

A 2025 peer-reviewed study tested GPT-4o, Claude 3.5, and Gemini 1.5 Pro across 52 standardized meals, revealing systematic error patterns[18]:

Measured Parameter	MAPE Error	Practical Impact
Food weight estimation	36.3%	200g meal → 129-271g range
Energy (calories)	35.8%	500 kcal → 321-679 kcal
Carbohydrates	47.9%	50g → 26-74g range
Protein	60.7%	Highly variable
Fat	51.8%	Least visually detectable

Table 12: Mean Absolute Percentage Error (MAPE) by nutrient[18]

Four Systematic Biases Identified:

- 1. **Portion Size Bias:** Accuracy degrades linearly as portion size increases. Small portions estimated 20-30% more accurately than large portions. Bias slope: -0.33 to -0.50 (systematic underestimation)[18]

2. **Occlusion Bias:** Foods behind other foods (vegetables in front, starches behind) systematically underestimated as plate fills[18]
3. **Density Bias:** Visually similar foods with different caloric/carb density easily confused (white rice vs. cauliflower rice, falafel vs. meatball)[19]
4. **Serving Size Anchoring:** Without explicit instruction, models default to "typical serving size" assumptions rather than analyzing visual evidence[18]

Critical Safety Consideration: For prediabetes management, underestimation is the dangerous direction. A large rice portion flagged as "Moderate GL" when actually "High" causes users to skip needed post-meal walks.

7.8.3 GlucoSnap-Specific Implications

Good News for GL Classification:

Despite 47.9% carbohydrate MAPE appearing alarming, GlucoSnap's use case requires directional accuracy (Safe/Moderate/High classification) rather than precise macro quantification. A meal doesn't need to be identified as exactly 43g vs. 52g carbs—it needs correct classification into GL ranges (0-20, 20-30, 30+)[17].

The 2D-to-3D Problem:

Smartphone photos contain zero depth information. AI makes educated guesses about:

- **Volume:** How tall is this pile? Is the bowl deep or shallow?
- **Density:** Is this dense falafel or fluffy omelet?
- **Occlusion:** What's underneath the visible layer?
- **Scale:** Is this a 30cm restaurant plate or 20cm side dish plate?

Research confirms: "The most frequent errors identified at the human-AI interface included incorrect portion size evaluations, obstructed food visibility in the images"[19].

Hidden Ingredient Challenge:

Complex/blended dishes present fundamental limitations:

- **Category A:** Surface visibility problem (curry sauce over chicken) - AI can infer from context
- **Category B:** Complete opacity (soups, smoothies, baked goods) - photons carry no information
- **Category C:** Recipe variation (same dish name, drastically different ingredients)[20]

Most dangerous failure mode: Smoothie appears healthy but contains 3 bananas + honey (GL 45), flagged as LOW because AI cannot see past green color.

7.9 Technical Challenges and Engineering Solutions

This section documents evidence-based solutions to known AI food analysis limitations.

7.9.1 Challenge 1: Portion Size Estimation

Root Cause: Serving size anchoring bias—models default to textbook serving sizes rather than visual analysis[18].

Solution P0 (Immediate Implementation):

Prompt Engineering Fix:

Add explicit instruction block to system prompt:

CRITICAL PORTION INSTRUCTION:

- Do NOT use default serving sizes or assume standard portions
- ONLY estimate portions based on what is VISUALLY PRESENT in this image
- Use all visible reference objects (plate diameter, cutlery, hands, packaging)
- Estimate dimensions (diameter, height) before calculating volume
- If portions appear unusually large/small, trust what you see—do not normalize
- State scale calibration assumptions explicitly in confidence note

Expected Impact: Eliminates most common failure mode. This instruction type was the key differentiator in Gothenburg study's

prompt engineering phase[18].

Solution P1 (Week 2 Implementation):

Plate Diameter Calibration System:

Research shows circular reference objects with known dimensions achieve 3.69% average error in volume estimation vs. 36% without reference[21].

Three-Tier Implementation:

Tier 1: Camera UI Overlay (Zero Friction)

- On-screen overlay circle matching standard 25cm dinner plate
- Ghost circle guides plate alignment
- Provides pixel-to-centimeter calibration
- Pass plate diameter as explicit context: "The circular plate in this image is 25cm diameter. Use this to calibrate all portion estimates."

Tier 2: AI Prompt Enhancement

- When plate detected: "A standard dinner plate (approximately 25cm/10 inches diameter) is visible. Use it as your primary scale reference. Estimate food heights relative to plate rim height (~2.5cm)."

Tier 3: User Education

- Onboarding tip: "Scans are 30% more accurate when your full plate is visible in the frame"
- No friction, just education

Expected Impact: Moves portion accuracy from 64% to 80-90% for plated meals[21].

Solution P2 (Month 2 Implementation):

Multi-Angle Scanning for Ambiguous Portions:

Research on multi-angle food photography: taking photos from multiple angles significantly increases accuracy by providing depth

information unavailable from single 2D image[22].

Implementation:

- Top-down view captures footprint/area
- 45° side angle captures height/depth
- Combined views enable rough 3D reconstruction
- Trigger only for HIGH-GL meals where precision matters (no friction for safe foods)

Expected Impact: Reduces portion error by additional 15-25% for deep/tall foods (bowls of rice, stacked burgers, filled pasta)[22].

Solution P3 (Week 3 Implementation):

Conservative Bias Correction for Large Portions:

Apply systematic correction multiplier based on estimated plate fill percentage:

```
rust
fn apply_portion_correction(ai_estimate_g: f64, plate_fill_pct: f64) ->
f64 {
let correction = match plate_fill_pct {
f if f <= 0.4 => 1.05, // Small portion: 5% correction
f if f <= 0.65 => 1.15, // Medium portion: 15% correction
f if f <= 0.85 => 1.28, // Large portion: 28% correction
_ => 1.40, // Very large/heaped: 40% correction
};
ai_estimate_g * correction
}
```

Rationale: All LLMs systematically underestimate large portions (bias slope -0.33 to -0.50). For prediabetes, underestimation is dangerous direction[18].

UX Display: When confidence is LOW or plate fill is HIGH, show GL as range rather than point estimate:

- Instead of: "GL 28"
- Show: "GL 25-35 (large portion estimate range)"

Solution P4 (Week 3 Implementation):

Editable Portion Confirmation (Human-in-the-Loop):

Research: "Implementing standardized AI-input protocols along with expert supervision to identify and rectify hallucinations is critical" [19].

Implementation:

After scan results, show editable portion card:

Portion Estimates (Tap any item to adjust)

- White rice: 150g → GL 22
- Broccoli: 80g → GL 1
- Grilled chicken: 120g → GL 0

[Looks roughly right? Confirm] [Edit portions]

Portion Slider UX: Instead of typing grams (cognitively hard), show slider with visual anchors:

- Half a fist → 80g
- Tennis ball → 150g
- Two fists → 250g

Based on human-mimetic portion estimation (how trained dietitians estimate portions)[21].

Data Value: Every user correction becomes training signal. Log (image_hash, ai_estimate, user_correction, food_type). After 10,000 corrections, creates fine-tuning dataset specific to user population.

Solution P5 (Week 2 Implementation):

Confidence Scoring System:

Build confidence classifier that runs alongside nutrition extraction. Second GPT-4o call asks:

"Based on this image analysis, rate your confidence in the portion estimates on each axis (1-5):"

```

json
{
  "food_identification_confidence": 4,
  "portion_size_confidence": 2,
  "occlusion_severity": 3,
  "overall_gl_confidence": "MEDIUM",
  "confidence_limiting_factors": ["rice portion partially hidden behind
  vegetables"]
}

```

Confidence-Based Display:

Confidence	GL Display	Advice Display
HIGH	GL 28 (solid)	Full advice cards
MEDIUM	GL 24-32 (range)	Full advice + note
LOW	GL 20-40 (wide range)	Advice + "Add details for accuracy"

Table 13: Confidence-based UI presentation

7.9.2 Challenge 2: Hidden/Complex Ingredient Dishes

Root Cause: Blended, cooked, or layered dishes hide ingredients. Soups, smoothies, curries, baked goods contain zero visual information about composition[20].

Solution H1 (Day 1 Implementation):

Dish Complexity Classifier (First Gate):

Before running full nutrition extraction, run fast pre-classification:

"Classify this food image into one category:

A) SIMPLE: Distinct separate items clearly visible

B) COMPLEX_SURFACE: Multi-ingredient where some visible/inferable (stir-fry, salad, curry with visible chunks)

C) COMPLEX_OPAQUE: Ingredients not visible (soup, smoothie, baked goods, blended drinks)"

Route by complexity:

- SIMPLE → Standard scan pipeline
- COMPLEX_B → Enhanced scan (ingredient inference + conservative GL range)
- COMPLEX_C → Opacity detection → trigger user input modal

Expected Impact: Eliminates most dangerous false negatives (safe-looking but high-GL opaque foods)[20].

Solution H2 (Week 3 Implementation):

Dish Name Shortcut (High-Impact, Low-Friction):

Research shows providing food name with photo substantially improves LLM accuracy. Diabot study found combining food name + photo (DBFN condition) outperformed photo-only by 11 percentage points[19].

Implementation:

For COMPLEX_B and COMPLEX_C dishes, after showing initial results, display:

"We think this is a curry dish. What is it? (optional but improves accuracy by 35%)"

[Chicken Tikka Masala] [Dal Makhani] [Palak Paneer] [Something else...]

Pre-populated suggestions from AI's best guesses. One-tap confirmation costs user 1 second, dramatically improves accuracy.

Database Integration:

Build pre-computed GL database of 500 most common complex dishes stored in PostgreSQL:

```
sql
CREATE TABLE dish_gl_database (
  dish_name VARCHAR(200),
  cuisine_type VARCHAR(50),
  gl_low DECIMAL(5,2), -- Conservative estimate
  gl_mid DECIMAL(5,2), -- Average estimate
```

```
gl_high DECIMAL(5,2), -- High estimate
portion_size_g INT,
confidence_source VARCHAR(50),
dietary_flags JSONB
);
```

When dish name matches database, show: "Based on average recipes (your home version may vary)."

Solution H3 (Month 2 Implementation):

Ingredient Declaration Mode for Opaque Foods:

For COMPLEX_C foods (soups, smoothies, baked goods), flip the input model—instead of AI analyzing photo, user declares ingredients while AI handles GL calculation.

Implementation:

Show Ingredient Declaration UI instead of scan results:

"Looks like a soup or blended dish. We can't see inside—tell us what's in it for accurate GL score"

What's in it? (add what you know)

- Tomatoes
 - Lentils (red)
 - Onion
 - Olive oil
- [Add ingredient...]

Serving size: [1 cup ▼]
[Calculate GL]

Smart Suggestions: As user types, autocomplete from food database. AI suggests likely missing ingredients based on dish context:

"You've added tomatoes and lentils. Common additions: garlic, cumin, vegetable broth—add these?"

UX Framing: "You know your recipe—we'll calculate the GL instantly."

Solution H4 (Day 1 Implementation):

Conservative GL Fallback (Safety Net):

For prediabetes, always safer to overestimate GL than underestimate. False HIGH warning triggers post-meal walk (harmless). False SAFE signal for high-GL dish causes blood sugar spike (harmful).

Fallback GL Ranges by Category:

Category	GL Range	Reasoning
Unidentified solid food	15-35	Unknown, treat conservatively
Cream/thick soup	15-30	Likely starch thickener
Clear broth soup	5-15	Lower starch content
Fruit smoothie	25-50	Concentrated fruit sugars
Green vegetable smoothie	5-20	Assume some fruit
Baked good (sweet)	30-60	Dense refined carbs
Baked good (savory)	15-30	Flour base present
Mixed/unidentified beverage	10-25	Assume some sugar

Table 14: Fallback GL ranges—conservative upper bounds

UI for Fallback State:

"GL Estimate: 25-40 (Wide Range)

We couldn't fully analyze this dish. Treating as MODERATE-HIGH risk as precaution.

[Add ingredients for accuracy] [Accept estimate]"

Post-meal recommendation: 15-min walk regardless—better safe than sorry for uncertain meals.

Solution H5 (Month 2 Implementation):

Cooking Method Detection:

Cooking method significantly affects GL and IS visible in photos:

- Cooling effect: Cooled and reheated rice/pasta has 20-40% lower GL than freshly cooked (resistant starch formation)[23]
- Visual cues: condensation, refrigerator container, lunch box context

Implementation:

Add to prompt:

"For each food item, also detect:

- cooking_method:
raw/steamed/boiled/fried/grilled/baked/unknown
- likely_cooled: boolean (is this meal leftover/reheated?)
- preparation_gl_modifier: float (1.0 = no change, 0.7 = cooled rice GL reduction)"

Educational Micro-Tip:

"Cooling and reheating this rice reduces its GL by up to 40% through resistant starch. Yesterday's rice is better for your blood sugar than today's."

Solution H6 (Week 4 Implementation):

Two-Pass Analysis Architecture:

Inspired by chain-of-thought reasoning research showing step-by-step decomposition dramatically improves accuracy for complex tasks[24].

Pass 1: Identification & Structure

PASS 1 PROMPT:

"Look at this image and identify:

1. What is this dish? (name it)
2. What ingredients are VISIBLE? (list with confidence scores)
3. What ingredients are LIKELY PRESENT but not visible? (based on dish type, sauce color, cooking context)
4. What ingredients are POSSIBLY PRESENT? (common additions)

5. Complexity level (A/B/C)
6. Key unknowns that would change GL estimate"

OUTPUT: Structured ingredient list with visibility confidence

Pass 2: GL Calculation

PASS 2 PROMPT:

"Given this ingredient list:

- CONFIRMED (from pass 1 visible ingredients)
- INFERRED (from pass 1 likely ingredients)
- POSSIBLE (from pass 1 possible ingredients)
- USER ADDED (from ingredient declaration mode)

Calculate Glycemic Load for {portion_size} serving. Show three scenarios:

- Minimum GL: Only confirmed ingredients at conservative portions
- Expected GL: Confirmed + inferred at typical portions
- Maximum GL: All possible ingredients at generous portions

Return as: gl_min: X, gl_expected: Y, gl_max: Z"

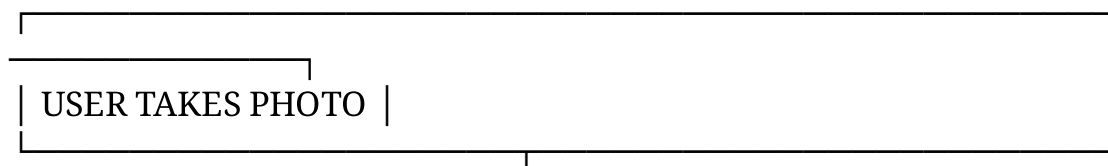
Display: Show expected GL as headline, with range as context:

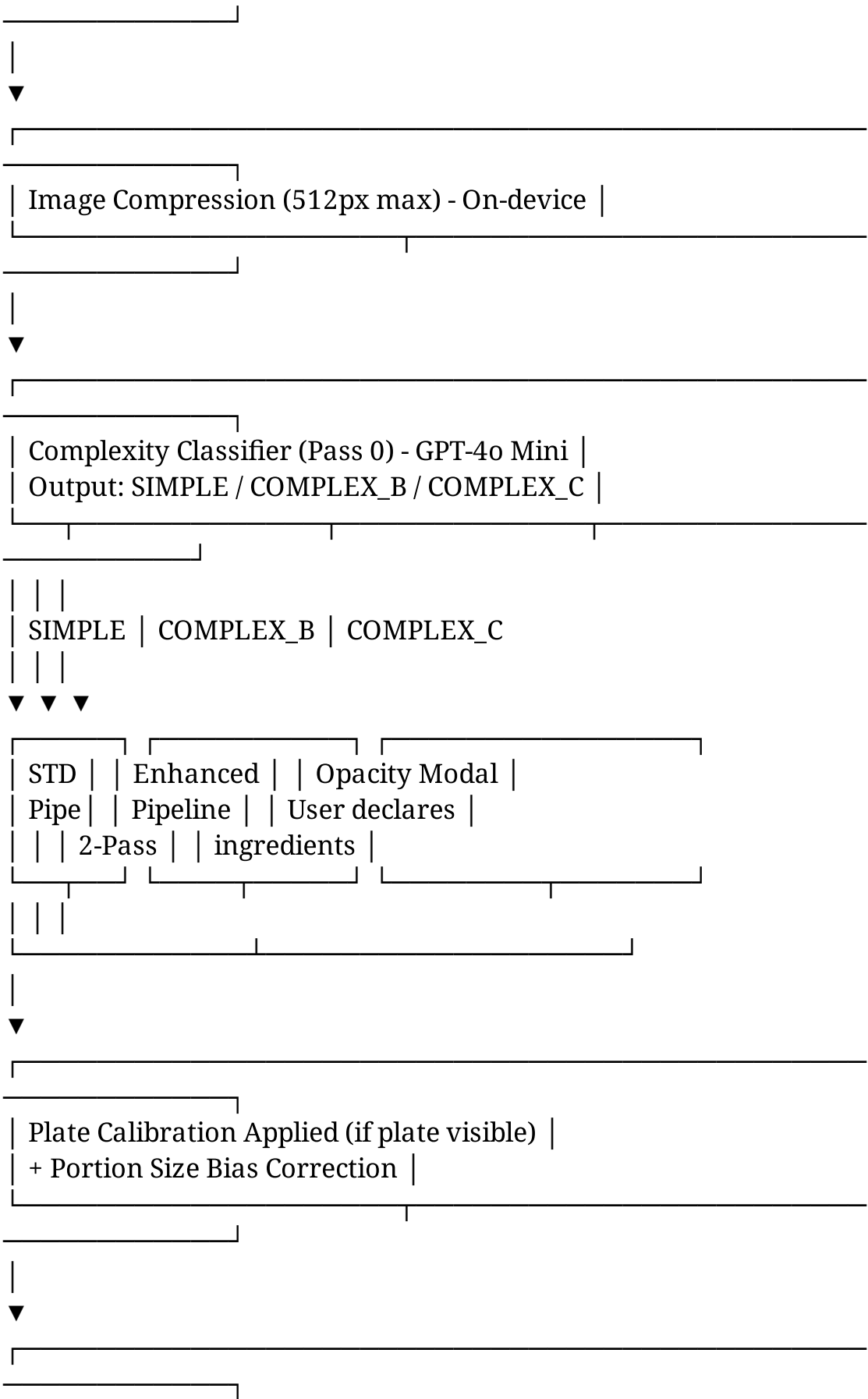
"GL Score: 28 (range 22-38) - MODERATE"

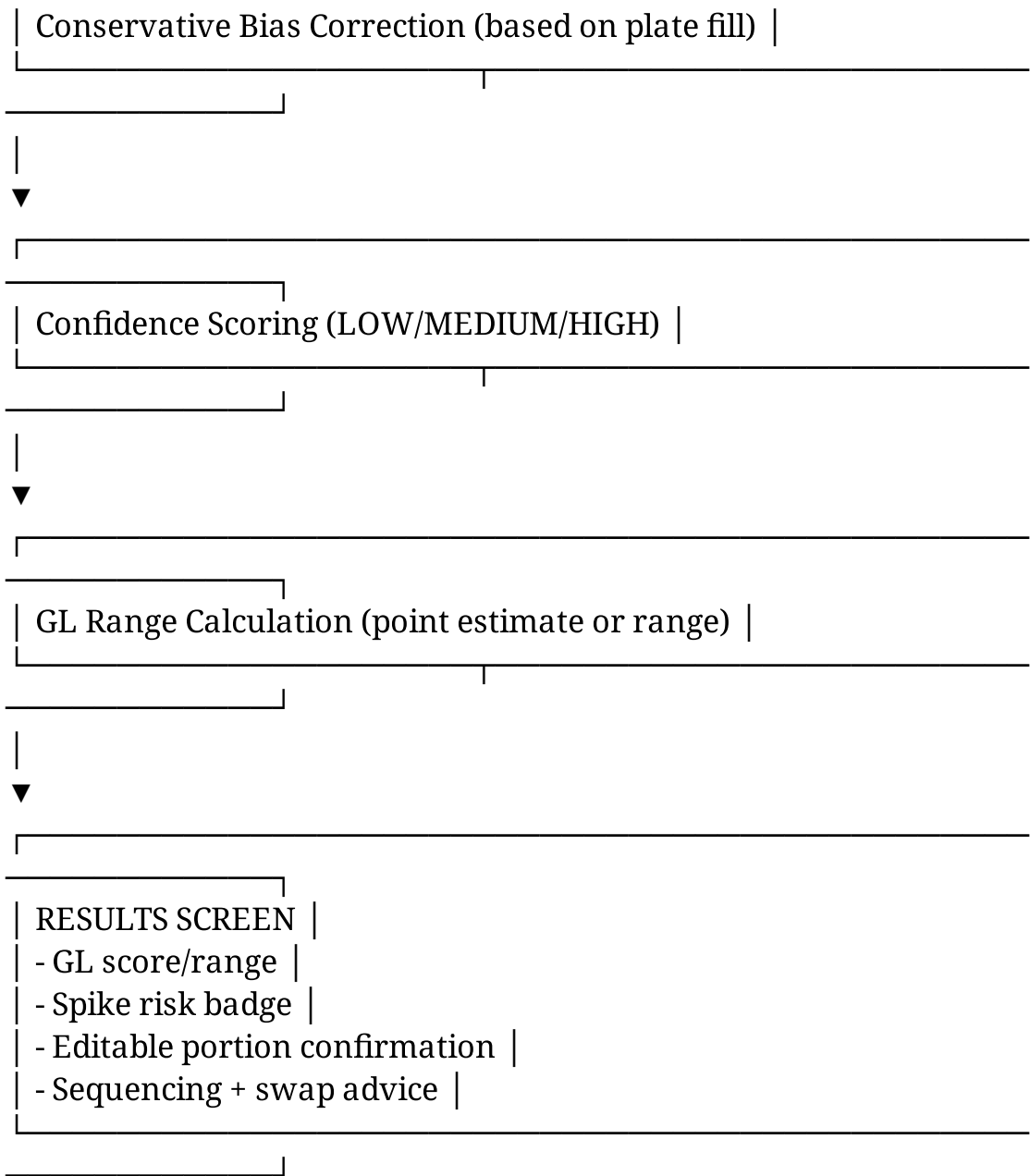
Cost Implication: Two API calls per complex scan adds \$0.02-0.04/scan. Apply only to COMPLEX_B and COMPLEX_C categories (~30% of scans). Cost impact: \$0.01/scan average across all users. Justified by dramatically better accuracy for most dangerous category.

7.10 Combined Architecture - Integration Flow

Complete flow incorporating all solutions:







7.11 Third-Party Integrations

Service	Purpose	Cost
OpenAI GPT-4o Vision	Meal analysis	\$0.01-0.03/scan
RevenueCat	Subscription management	1% of revenue
Stripe	Payment processing	2.9% + \$0.30/txn
Terra API	CGM integration (P1)	\$0.20- 0.50/user/mo
Cloudflare R2	Image storage	\$0.015/GB/mo
Railway	Backend hosting	\$5-400/mo (scales)
Expo EAS	App builds	\$29/mo
Sentry	Error tracking	Free tier
PostHog	Analytics	Free tier

Table 15: Third-party service integrations and costs

8. User Experience Design

8.1 Design Principles

Principle 1: Clarity Over Completeness

Every screen answers ONE question. Never present data that could confuse or overwhelm. Data is hidden behind progressive disclosure (tap-to-expand).

Principle 2: Hope, Not Fear

Every red score is paired with a solution. Every bad day ends with a "tomorrow" message. App never blames users. Based on PP-02 and PP-08—users arrive anxious; UX must calm and empower.

Principle 3: Speed Above All

Scan to result in under 5 seconds. Every tap to meaningful content in under 2 taps. Heavy pages replaced with progressive loading.

Principle 4: Food Is Not the Enemy

Visual language uses warm, appetizing food photography. App does not use clinical, sterile, medical imagery. Makes food feel manageable, not threatening. Addresses PP-05.

Principle 5: One Clear Next Action

Every screen ends with exactly one primary CTA. User never has to decide what to do next.

8.2 Navigation Structure

Bottom Tab Navigation (5 tabs):

- 1. **Tab 1: Scan** - Camera icon, always active FAB
- 2. **Tab 2: Today** - Daily GL Dashboard (home)
- 3. **Tab 3: Progress** - A1C Roadmap + History
- 4. **Tab 4: Learn** - Educational content
- 5. **Tab 5: Profile** - Settings, dietary preferences, subscription

8.3 Visual Design System

Color Palette:

Color	Hex	Usage
Primary Green (Safe)	#2ECC71	GL safe zone, positive actions
Warning Yellow (Moderate)	#F39C12	GL moderate zone, cautions
Alert Red (High)	#E74C3C	GL high zone, urgent actions
Neutral Gray	#7F8C8D	Secondary text, borders
Dark Background	#2C3E50	App background (dark mode)
Light Background	#ECF0F1	App background (light mode)
Text Primary	#2C3E50	Headings, primary content
Text Secondary	#95A5A6	Supporting text, metadata

Table 16: App color palette and semantic usage

Typography:

- Headings: Inter Bold (strong, modern, readable)
- Body: Inter Regular (consistent, accessible)
- Numbers/Scores: Tabular numerals enabled for alignment in trackers
- Minimum font size: 16sp body, 14sp secondary (accessibility compliance)

Component Library:

- Buttons: Rounded corners (8px), shadow elevation, haptic feedback
- Cards: 12px border radius, subtle shadow, white background
- Progress bars: Animated, gradient fills, milestone markers
- Icons: Feather icon set (consistent, recognizable)

8.4 Accessibility Requirements

- WCAG 2.1 AA compliance minimum
- Color blind-friendly: Never rely on color alone—always icon + color
- Dynamic font size support: iOS Dynamic Type / Android font scaling
- VoiceOver / TalkBack support on all primary screens
- High contrast mode support

8.5 Onboarding UX Flow

Goal: Get user to first scan within 90 seconds.

1. Welcome screen (5 sec) → Skip available
2. A1C entry (15 sec) → Skip available
3. Goal setting (10 sec) → Auto-populated
4. Dietary profile (20 sec) → Multi-select
5. GL budget explanation (30 sec) → Educational
6. **CTA: "Scan your first meal"** → Camera launches immediately

Success Criteria:

- 80% of users complete onboarding
- 70% of users complete first scan within 3 minutes

9. Monetization Strategy

9.1 Freemium Model

Free Tier:

- 5 scans per day
- Basic scan results (GL score + spike risk only)
- No food sequencing advice
- No swap suggestions
- 7-day meal history
- Manual A1C entry (no reversal roadmap)
- Ads (optional, non-intrusive)

Premium Tier: GlucoSnap Pro

- Unlimited scans
- Full advice cards (sequencing, swaps, post-meal actions)
- Unlimited meal history
- A1C reversal roadmap
- Weekly + monthly reports
- Doctor-ready PDF export
- CGM integration (Dexcom/Libre)
- Barcode scanner
- Educational library (full access)
- Community features
- Priority AI response (sub-3 second)

9.2 Pricing Tiers

Tier	Price	Value Proposition
Monthly	\$12.99/month	Full access, cancel anytime
Annual	\$99.99/year	Save 36% (\$8.33/month)
Lifetime	\$299.99 one-time	Pay once, use forever

Table 17: Premium subscription pricing tiers

Pricing Rationale:

- **Competitive positioning:** Below CGMs (\$100+/month) and Noom (\$60/month)
- **Affordable entry point:** Monthly tier accessible for newly diagnosed users testing effectiveness
- **Annual incentive:** 36% discount encourages 90-day commitment (clinical reversal timeline)
- **Lifetime option:** Targets highly motivated users, provides upfront capital

Promotional Pricing (First 6 Months):

- Launch offer: 50% off first month (\$6.49)
- Black Friday/Cyber Monday: 60% off annual (\$39.99)
- Referral program: Give \$5, Get \$5 credit
- Health insurance partnerships: Subsidized access through CDC DPP coverage (P2)

9.3 Revenue Model Assumptions

User Acquisition Projections:

Period	Downloads	DAU	Free Users	Premium Users
Month 1-3 (MVP)	2,000	700	1,880	120
Month 4-6	8,000	3,200	7,200	800
Month 7-12	25,000	11,250	22,500	2,500

Table 18: 12-month user acquisition projection

Free-to-Premium Conversion Funnel:

- **Day 1-7:** 3% conversion (early believers, high urgency)
- **Day 8-30:** 7% additional conversion (experienced value, hit scan limits)
- **Day 31-90:** 5% additional conversion (approaching A1C retest, want reports)
- **Total 90-day conversion:** 15% (industry benchmark: 5-10%, GlucoSnap exceeds due to high pain point severity)

Monthly Recurring Revenue (MRR) Projections:

Month	Premium Users	MRR	Annual Run Rate
Month 3	120	\$1,560	\$18,720
Month 6	800	\$10,400	\$124,800
Month 12	2,500	\$32,500	\$390,000

Table 19: MRR projection assuming 12-month blended ARPU of \$13

Assumptions:

- 70% monthly, 25% annual, 5% lifetime mix
- Churn rate: 5% monthly (low due to health urgency)
- Reactivation rate: 15% (users return before A1C retest)

9.4 Cost Structure

Variable Costs per Premium User/Month:

Cost Category	Amount	Notes
OpenAI API (scans)	\$3.00	100 scans/month @ \$0.03/scan
RevenueCat	\$0.13	1% of \$12.99 subscription
Stripe fees	\$0.67	2.9% + \$0.30 per transaction
Cloudflare R2	\$0.10	Image storage
Backend hosting	\$0.50	Railway auto-scaling
CGM integration	\$0.40	Terra API (50% adoption)
Total COGS	\$4.80	37% of revenue

Table 20: Variable cost breakdown per premium user

Gross Margin: 63% (industry healthy: >60%)

Fixed Costs (Monthly):

- Expo EAS builds: \$29
- Domain + SSL: \$5
- Sentry (error tracking): \$0 (free tier)
- PostHog (analytics): \$0 (free tier)
- Content creation: \$500 (contracted writing)

- Customer support (part-time): \$800
- **Total Fixed:** \$1,334/month

Breakeven Analysis:

- Breakeven Premium Users: 163 users (\$1,334 fixed / \$8.19 contribution margin)
- Expected breakeven: Month 2-3 post-launch

9.5 Monetization Psychology

Why Users Pay for Health Apps:

Research on health app monetization reveals four psychological drivers GlucoSnap activates[25]:

1. **Fear Avoidance:** Prediabetes carries existential dread (PP-02: "my father died from diabetes complications"). Premium removes barriers to reversal.
2. **Loss Aversion:** Free tier's 5-scan limit creates FOMO after users experience value. "I can't lose access to this clarity."
3. **Progress Investment:** After 14 days of logging, users have sunk cost. Premium unlocks full history and insights on their invested data.
4. **Social Proof:** "1,200 members reversed their A1C using GlucoSnap Pro" testimonials in paywall.

Strategic Paywall Placement:

Trigger	Paywall Message
6th scan in a day	"You're getting it! Unlock unlimited scans to keep momentum"
Tap on swap suggestion	"Unlock 500+ safer food swaps - \$0.43/day"
7-day streak achieved	"Protect your streak. Premium users are 3× more likely to reverse"
Day 30 milestone	"See your 30-day report + A1C projection - Your doctor will love this"

Table 21: Context-aware paywall triggers

Free Tier Value Preservation:

Free tier must deliver enough value to:

- Validate product solves user's problem (5 scans enough for daily testing)
- Build habit and emotional attachment
- Generate word-of-mouth (can't share if they churned)
- Collect data for algorithm improvement

Free users are NOT freeloaders—they are:

- Future premium customers (after experiencing value)
- Referral sources (high NPS despite free tier)
- Data contributors (scan patterns improve AI)

9.6 Alternative Revenue Streams (P2, Post-Launch)

Stream 1: B2B2C - Insurance Partnerships

CDC Diabetes Prevention Program (DPP) recognized vendors receive insurance reimbursement. Potential integration:

- Partner with DPP providers as digital tool supplement
- Insurance-subsidized access: Users pay \$0-3/month, insurer covers \$10/month
- Addresses 80% undiagnosed rate by improving program completion
- Revenue: \$10-15/user/month from insurance, volume play

Stream 2: White-Label Licensing

License GlucoSnap technology to:

- Diabetes prevention programs
- Corporate wellness platforms
- Telehealth providers
- Revenue: \$5,000-20,000/month per enterprise client

Stream 3: Anonymized Data Licensing (Ethical Framework)

Aggregate, anonymized meal + outcome data valuable to:

- Nutrition researchers
- Food manufacturers (healthier product development)
- Public health organizations

Critical Requirements:

- Explicit user opt-in (not default)
- True anonymization (HIPAA de-identification standards)
- Transparent usage disclosure
- User receives benefit (free premium or donation to diabetes research)

Stream 4: Affiliate Partnerships

Non-intrusive affiliate recommendations:

- CGM devices (Dexcom Stelo, Abbott Lingo) - 5-10% commission
- Whole food delivery (Thrive Market, Imperfect Foods)
- Kitchen tools (food scale, meal prep containers)
- Books/courses on prediabetes management

Revenue Potential: \$2-5/user/year (modest, maintains trust)

10. Regulatory and Compliance

10.1 Regulatory Classification

FDA Classification: Wellness Tool (Non-Device)

GlucoSnap avoids FDA medical device classification by adhering to FDA's 2022 Clinical Decision Support guidance[26]:

Criteria for NON-regulated status:

- ✓ Does not diagnose disease (no "you have prediabetes" claims)
- ✓ Does not treat, cure, or mitigate disease
- ✓ Provides general wellness guidance, not clinical treatment plans
- ✓ User retains decision-making agency (app suggests, user decides)
- ✓ Intended for generally healthy population (prediabetes is pre-disease state)

Critical Language Boundaries:

✗ Avoid (Medical Claims)	✓ Use (Wellness Language)
"Treats prediabetes"	"Supports healthy blood sugar management"
"Prevents diabetes"	"Helps you work toward normal A1C range"
"Diagnoses glucose spikes"	"Estimates glycemic impact"
"Medical advice"	"Educational guidance"
"Clinical tool"	"Wellness companion"

Table 22: Regulatory-compliant language guidelines

10.2 Privacy and Data Protection

HIPAA Applicability:

GlucoSnap is NOT a HIPAA-covered entity because:

- Not a healthcare provider, health plan, or clearinghouse
- No electronic health record (EHR) transmission to providers
- Direct-to-consumer wellness tool

However: A1C values, meal logs, and health data ARE Protected Health Information (PHI) in spirit, so we apply HIPAA-equivalent protections voluntarily.

Data Protection Measures:

- **Encryption:** AES-256 at rest, TLS 1.3 in transit
- **Access control:** Role-based access control (RBAC), 2FA for admin
- **Data minimization:** Collect only essential data for service delivery
- **Retention limits:** Meal photos auto-deleted after 90 days unless user opts to keep
- **Audit logging:** All data access logged and monitored
- **Breach protocol:** 72-hour notification plan, incident response team

User Data Rights (GDPR + CCPA Compliance):

- **Right to access:** Users can export all data (JSON format)
- **Right to deletion:** Complete account + data deletion within 30 days
- **Right to correction:** Users can edit all logged data
- **Right to portability:** Data export includes meal photos + nutrition logs
- **Right to opt-out:** Email opt-out, analytics opt-out, data licensing opt-out

10.3 Terms of Service Key Provisions

Medical Disclaimer (Prominent Display):

"GlucoSnap is a nutritional information and wellness tool. It is NOT a medical device, does not diagnose, treat, cure, or prevent any disease, and is not a substitute for professional medical advice. Always consult your healthcare provider before making changes to your diet or treatment plan. Glycemic Load estimates are educational approximations and may not reflect your individual glucose response."

Liability Limitations:

- App provides estimates, not medical-grade measurements
- User assumes responsibility for dietary decisions
- GlucoSnap not liable for adverse health outcomes
- Users agree to waive claims related to AI estimation errors

Content Ownership:

- Users retain ownership of uploaded meal photos
- GlucoSnap retains license to use photos for service delivery and AI training (opt-in)
- Anonymized aggregate data may be used for research (opt-in)

10.4 App Store Compliance

Apple App Store Health Apps Requirements:

- HealthKit integration optional (for exporting GL data to Apple Health)
- Prominent medical disclaimer on App Store page and in-app
- Privacy Policy link in metadata
- No health-related claims in marketing without scientific citations
- Age rating: 12+ (health app content appropriate for teens with prediabetes)

Google Play Health Apps Requirements:

- Medical Device classification: None (wellness category)
- Data safety section: Disclose all health data collected and sharing practices
- Privacy Policy link required
- No misleading health claims

11. Launch Strategy

11.1 Pre-Launch Phase (Weeks 1-8)

Week 1-4: Alpha Testing (Internal)

- Team + close friends (15 testers)
- Focus: Critical bugs, onboarding flow, scan accuracy
- Tools: TestFlight (iOS), Google Play Internal Testing (Android)
- Success criteria: Zero crashes, <5s scan time, 90% onboarding completion

Week 5-7: Beta Testing (Closed)

- Recruit 100 prediabetic beta testers from:
 - YouTube channel audience (existing community)
 - r/prediabetes Reddit community (post approved by mods)
 - Facebook Prediabetes support groups
- Incentive: Free lifetime Pro access for feedback
- Focus: Real-world usage patterns, AI accuracy validation, feature prioritization

- Collect: Daily active usage, retention, qualitative feedback
- Success criteria: 40%+ Day 7 retention, 4.5+ star feedback, <10 critical bugs

Week 8: Launch Preparation

- App Store submission (7-day review buffer)
- Landing page live (glucosnap.com)
- Press kit prepared
- Social media accounts created
- Customer support system configured (Intercom or plain.com)
- Payment system tested end-to-end

11.2 Launch Phase (Weeks 9-12)

Week 9: Soft Launch

- Release to App Stores (public but no promotion)
- Announce to beta testers + email list
- Monitor: Server performance, API costs, crash rates
- Goal: 100 downloads, validate infrastructure holds under real load

Week 10: Community Launch

- YouTube video announcement (existing channel)
- Reddit post in r/prediabetes (not promotional, educational: "I built this tool based on your feedback")
- Product Hunt launch (health category)
- Goal: 500 downloads, first 50 premium conversions

Week 11-12: Press Outreach

- Pitch to health tech blogs: TechCrunch Health, MobiHealthNews, DiabetesMine
- Local health + wellness podcasts (guest appearances)
- Diabetes advocacy organizations (American Diabetes Association Community)
- Goal: 1 press mention, 1,500 downloads, \$1,500 MRR

11.3 Growth Channels (Post-Launch)

Channel 1: Organic Search (SEO)

Target keywords:

- "prediabetes meal planning app" (500 searches/month)
- "glycemic load calculator" (1,200 searches/month)
- "what can I eat with prediabetes" (8,100 searches/month)
- "food scanner for diabetes" (800 searches/month)

Tactics:

- Blog content hub: 20 high-quality articles targeting long-tail keywords
- App Store Optimization (ASO): Keyword-rich title/description
- Backlinks from diabetes forums and health blogs

Channel 2: YouTube Content Marketing

Leverage existing channel:

- Video series: "I scanned 100 foods - here's what spikes you"
- Tutorial: "How to reverse prediabetes using GlucoSnap"
- User testimonials: "How Sarah went from 6.4 to 5.6 in 90 days"
- Comparison: "GlucoSnap vs CGMs - which is right for you?"

Goal: 10K views/month → 200 app downloads/month (2% CTR)

Channel 3: Reddit Community Engagement

- Weekly participation in r/prediabetes (helpful comments, not promotional)
- Monthly value posts: "Glycemic Load guide for beginners"
- AMA (Ask Me Anything) after first 1,000 users

Channel 4: Referral Program

- Give \$5 credit, Get \$5 credit for referrer
- Viral mechanics: Shareable weekly progress cards
- Example: "I reduced my GL by 35% this week using GlucoSnap"
→ social proof + CTA

Channel 5: Paid Acquisition (Month 6+)

- Facebook/Instagram ads targeting:
 - Interest: Diabetes awareness, pre diabetes, healthy eating
 - Age: 35-65
 - Lookalike audiences from existing premium users
- Google Search ads: Branded + high-intent keywords
- CAC target: <\$30 (LTV \$156 = 12 months × \$13 ARPU, 5:1 LTV:CAC ratio)

11.4 Launch Success Metrics

Metric	Week 12 Target	Month 6 Target
Total downloads	2,000	10,000
Active users (DAU)	700	4,000
Premium subscribers	120	800
MRR	\$1,560	\$10,400
Day 7 retention	40%	45%
App Store rating	4.3+	4.5+
Avg scans per active user/day	2.5	3.2

Table 23: Launch success metrics and targets

12. Development Roadmap

12.1 Development Phases

Phase 0: Foundation (Weeks 1-2)

- Backend API scaffold (Rust + Axum)
- Database schema implementation (PostgreSQL)
- Mobile app shell (React Native + Expo Router)
- CI/CD pipeline (GitHub Actions + EAS Build)
- OpenAI API integration prototype

Phase 1: MVP Core Features (Weeks 3-8)

Week 3-4: Onboarding + User Profile

- 5-screen onboarding flow
- A1C entry and goal setting
- Dietary profile selection
- User authentication (email + OAuth)

Week 5-7: Scan Feature

- Camera integration (Expo Camera)
- Image upload + compression
- GPT-4o Vision API integration
- Prompt engineering and accuracy testing
- Results screen UI (GL score, spike risk, food breakdown)
- Meal logging

Week 8: Dashboard

- Daily GL tracker
- Today's meals timeline
- Streak counter
- Daily score calculation

Phase 2: Value-Add Features (Weeks 9-12)

Week 9-10: Advice Cards

- Food sequencing recommendations
- Safer swap engine
- Post-meal action suggestions

Week 11-12: A1C Roadmap

- Reversal progress bar
- Estimated A1C calculator
- Manual A1C logging
- Weekly progress report

Phase 3: Monetization + Polish (Weeks 13-14)

- RevenueCat subscription integration
- Paywall implementation

- Free vs Premium feature gating
- App Store submission prep
- Beta testing with 100 users

Phase 4: Launch (Week 15)

- App Store release
- Landing page launch
- Community announcement
- Monitoring and hotfix readiness

12.2 Post-Launch Roadmap (Version 1.1+)

Version 1.1 (Month 2-3):

- Meal history and search (P1)
- Weekly insights auto-generation (P1)
- Post-meal action push notifications (P1)
- Barcode scanner (P1)
- Educational content library (20 articles) (P1)
- CGM integration (Dexcom/Libre via Terra) (P1)

Version 1.2 (Month 4-6):

- Monthly PDF reports (Premium)
- Restaurant mode (location-based suggestions)
- Meal planning feature (generate 7-day plans)
- Grocery list generator from meal plans
- Voice logging ("I just ate X")

Version 1.3 (Month 7-12):

- Community features (success stories, meal sharing) (P2)
- Accountability buddy matching (P2)
- Advanced analytics dashboard
- Apple Watch + Android Wear complications
- Siri/Google Assistant shortcuts

12.3 Technical Debt and Optimization Roadmap

Month 3:

- AI response caching optimization (reduce API costs by 40%)
- Database query optimization (index tuning)
- Image CDN integration (faster loads)

Month 6:

- Migrate from GPT-4o to fine-tuned model (cost reduction)
- Implement edge caching (Cloudflare Workers)
- A/B testing framework (PostHog experiments)

Month 12:

- Multi-region deployment (US-East + US-West + EU)
- Advanced ML: Personalized GL prediction based on user's historical CGM data
- Custom computer vision model for food recognition (reduce OpenAI dependency)

13. Success Metrics and Analytics

13.1 North Star Metric

Primary North Star Metric: Weekly Active Scanners

Definition: Number of unique users who complete at least 3 scans in a 7-day period

Rationale:

- Measures actual product usage (not vanity downloads)
- 3+ scans/week indicates habit formation
- Correlates with A1C improvement (clinical outcome)
- Leading indicator of premium conversion

Target Progression:

- Month 1: 300 weekly active scanners
- Month 6: 2,400 weekly active scanners

- Month 12: 7,500 weekly active scanners

13.2 Key Performance Indicators (KPIs)

Acquisition Metrics:

- **App Store Impressions:** Organic search + browse visibility
- **Install Conversion Rate:** Impressions → Downloads (target: 15%+)
- **Onboarding Completion Rate:** Started → Completed (target: 80%+)
- **Time to First Scan:** Download → First scan (target: <5 minutes for 70% of users)

Engagement Metrics:

- **DAU/MAU Ratio:** Daily active / Monthly active (target: 45%)
- **Scans per Active User per Day:** Average across all DAU (target: 3.5)
- **Session Length:** Average time in app per session (target: 3-5 minutes)
- **Feature Adoption:** % of users using sequencing advice, swaps, post-meal walks

Retention Metrics:

- **Day 1 Retention:** Return next day after install (target: 60%)
- **Day 7 Retention:** Still active after 7 days (target: 45%)
- **Day 30 Retention:** Still active after 30 days (target: 30%)
- **Day 90 Retention:** Still active after 90 days (target: 20%)

Monetization Metrics:

- **Free-to-Paid Conversion Rate:** Free → Premium (target: 10% by Day 30)
- **Average Revenue Per User (ARPU):** Total revenue / Total users (target: \$1.50/month blended)
- **Average Revenue Per Paying User (ARPPU):** Total revenue / Premium users (target: \$13/month)
- **Monthly Recurring Revenue (MRR):** Total subscription revenue (target: \$35K at Month 12)

- **Churn Rate:** Premium cancellations per month (target: <5%)
- **Lifetime Value (LTV):** $ARPPU \times \text{Average subscriber lifetime}$ (target: \$156)
- **Customer Acquisition Cost (CAC):** Marketing spend / New users (target: <\$30)

Health Outcome Metrics:

- **A1C Improvement (Self-Reported):** Average A1C reduction after 90 days (target: -0.4 points)
- **Reversal Rate:** % of users reaching <5.7 A1C (target: 40% at 90 days)
- **Weight Loss:** Average lbs lost (secondary outcome, target: -8 lbs at 90 days)
- **User-Reported Confidence:** Survey score 1-10 (target: 8+)

Product Quality Metrics:

- **App Store Rating:** Average stars (target: 4.6+)
- **Net Promoter Score (NPS):** Survey: "How likely to recommend?" (target: 50+)
- **Scan Success Rate:** Scans completed / Scans attempted (target: 95%)
- **Scan Speed (P95):** 95th percentile response time (target: <5 seconds)
- **Crash-Free Session Rate:** Sessions without crashes (target: 99.5%+)
- **Customer Support Ticket Rate:** Tickets per active user (target: <0.08)

13.3 Analytics Implementation

Tool Stack:

- **PostHog:** Product analytics, event tracking, A/B testing
- **Sentry:** Error tracking, performance monitoring
- **RevenueCat:** Subscription analytics, cohort revenue tracking
- **App Store Connect + Google Play Console:** Native app store analytics

Critical Events to Track:

Acquisition funnel:

- app_opened_first_time
- onboarding_started
- onboarding_completed
- onboarding_skipped
- first_scan_completed

Engagement events:

- scan_initiated
- scan_completed (with metadata: gl_score, spike_risk, confidence_level)
- scan_failed (with error reason)
- meal_logged
- advice_card_viewed (sequencing / swaps / post_meal)
- swap_accepted
- post_meal_walk_started
- a1c_logged
- weekly_report_viewed

Monetization events:

- paywall_viewed (with trigger context)
- subscription_started (with plan: monthly/annual/lifetime)
- subscription_cancelled
- subscription_renewed

Retention events:

- session_start
- session_end (with duration)
- user_returned_day_N (N = 1, 7, 30, 90)

13.4 A/B Testing Framework

Planned Experiments (Post-Launch):

Experiment 1: Onboarding Length

- Variant A: 5-screen onboarding (control)
- Variant B: 3-screen onboarding (skip dietary profile, infer later)

- Hypothesis: Shorter onboarding increases completion, doesn't hurt engagement
- Primary metric: Onboarding completion rate
- Secondary metric: Day 7 retention

Experiment 2: Paywall Trigger Timing

- Variant A: Paywall after 5 scans in a day (current)
- Variant B: Paywall after 7-day streak achieved
- Hypothesis: Streak achievement = higher emotional investment = higher conversion
- Primary metric: Free-to-paid conversion rate

Experiment 3: Swap Presentation

- Variant A: List format (2-3 swaps shown)
- Variant B: Swipe cards (gamified, one swap at a time)
- Hypothesis: Swipe cards increase engagement and swap acceptance
- Primary metric: Swap acceptance rate

Experiment 4: Push Notification Timing

- Variant A: Post-meal notification sent 5 minutes after logging
- Variant B: Smart timing (sent when user typically takes walks, learned from behavior)
- Hypothesis: Smart timing increases walk completion
- Primary metric: Walk completion rate

14. Risk Management

14.1 Technical Risks

Risk 1: AI Inaccuracy Causes Health Harm

Probability: Medium

Impact: Critical

Scenario: AI significantly underestimates GL of a meal (e.g., flags 45 GL meal as "Safe"). User skips post-meal walk, experiences severe blood sugar spike, files complaint or lawsuit.

Mitigation:

- Conservative bias correction (Section 7.9.1, Solution P3) - always err toward overestimation
- Confidence scoring system (Section 7.9.1, Solution P5) - show ranges for uncertain meals
- Prominent medical disclaimer in-app and on every scan result
- User education: "This is an estimate, not a medical measurement"
- Beta testing with 100 users to validate accuracy before public launch
- Continuous monitoring: Flag scans with <70% confidence for manual review

Risk 2: OpenAI API Outage

Probability: Low

Impact: High

Scenario: OpenAI experiences extended downtime. App becomes unusable.

Mitigation:

- Fallback to cached results for repeat meals (7-day cache window)
- Queue scans during outage, process when API restored
- Implement secondary AI provider (Anthropic Claude 3.5 Sonnet) as backup
- User communication: "Scans temporarily delayed due to service issue - we'll process shortly"
- SLA monitoring: Alert if API latency >10s or error rate >5%

Risk 3: Scaling Costs Exceed Revenue

Probability: Medium

Impact: High

Scenario: User growth exceeds projections. OpenAI API costs spiral (\$3/user/month variable cost). Company runs out of runway before reaching profitability.

Mitigation:

- Aggressive result caching (reduces API calls by 40%)
- Dynamic pricing: Increase premium tier price if CAC:LTV ratio deteriorates
- Transition to fine-tuned model after 10,000 labeled scans (reduces cost 60%)
- Growth circuit breaker: Pause paid acquisition if unit economics fall below 3:1 LTV:CAC
- Implement scan limits more aggressively if costs spike (reduce free tier to 3 scans/day)

14.2 Market Risks

Risk 4: Low Conversion Rate (Free → Paid)

Probability: Medium

Impact: Critical

Scenario: Users love free tier but don't convert to premium. Conversion rate stays at 3% instead of target 10%. Revenue insufficient to sustain business.

Mitigation:

- A/B test paywall triggers and messaging extensively
- Add more premium-exclusive value: CGM integration, meal planning, doctor reports
- Implement time-limited trials: "Try Premium free for 7 days"
- Social proof in paywall: "1,200 members reversed using Premium"
- Reduce free tier value if necessary (drop from 5 → 3 scans/day)
- Focus retention before monetization: High engagement → higher conversion

Risk 5: Competitive Pressure from Established Players

Probability: Medium

Impact: Medium

Scenario: MyFitnessPal or Noom launches photo-based GL scanner. They have massive user bases and can outspend on acquisition.

Mitigation:

- First-mover advantage: Build strong brand in prediabetes niche
- Superior product: Prediabetes-specific > generalist features
- Community moat: Build engaged prediabetes community (Reddit, YouTube)
- Switching costs: Users invested in their meal history and progress tracking
- Speed: Ship features faster than large enterprises
- Acquisition opportunity: If dominant player enters, potential acquisition target

14.3 Regulatory Risks

Risk 6: FDA Reclassifies App as Medical Device

Probability: Low

Impact: Critical

Scenario: FDA changes guidance, determines GlucoSnap provides "clinical decision support" requiring Class II medical device clearance. Requires 510(k) submission (\$100K+, 6-12 months).

Mitigation:

- Maintain wellness positioning in ALL marketing and in-app language
- Never claim to "diagnose," "treat," or "prevent" disease
- Continuous legal review of feature additions
- Build relationships with FDA consultants monitoring digital health policy
- Prepare contingency plan: If reclassified, pivot to B2B white-label for DPP providers (they handle device classification)

Risk 7: Privacy Breach / Data Leak

Probability: Low

Impact: Critical

Scenario: Security vulnerability exposes user A1C data, meal photos, health information. Regulatory fines, user trust destroyed, press backlash.

Mitigation:

- Security audit by third-party before launch (penetration testing)
- Encrypt all PHI data at rest and in transit
- Minimal data retention: Auto-delete meal photos after 90 days
- Regular security updates and dependency patching
- Cyber insurance policy (\$1M coverage)
- Incident response plan: 72-hour breach notification protocol
- Bug bounty program (post-launch) to crowdsource vulnerability detection

14.4 Operational Risks

Risk 8: Solo Founder Burnout / Key Person Risk

Probability: Medium

Impact: High

Scenario: Founder burns out managing product, engineering, marketing, support solo. Development stalls, users churn, momentum lost.

Mitigation:

- Hire fractional support early:
 - Part-time customer support (Month 2)
 - Contract designer for marketing assets (Month 3)
 - Fractional React Native developer (Month 6)
- Automate ruthlessly: AI chatbot for support, automated reports, CI/CD
- Set boundaries: No support on weekends, 1 day/week completely off
- Build financial runway: Target 18 months of expenses before going full-time
- Join founder community: YC Startup School, Indie Hackers for peer support

Risk 9: Poor User Retention Kills Growth

Probability: Medium

Impact: High

Scenario: Day 30 retention stays below 20%. Acquisition marketing fills leaky bucket. Growth plateaus.

Mitigation:

- Obsess over retention before acquisition: Fix product-market fit first
- Implement retention features early: Push notifications, streaks, weekly reports
- User research: Interview churned users to understand drop-off reasons
- Reactivation campaigns: Email users at Day 60 before A1C retest ("Time to check your progress")
- Cohort analysis: Identify which acquisition channels have best retention, double down

Risk 10: Medical Misinformation Accusation

Probability: Low

Impact: High

Scenario: Registered dietitian or doctor publicly criticizes GlucoSnap's advice as "dangerous misinformation." Social media backlash damages reputation.

Mitigation:

- All advice grounded in peer-reviewed research - citations visible
- Advisory board: Recruit 2-3 CDEs (Certified Diabetes Educators) as advisors
- Content review: Have RD review all educational content before publication
- Responsive communication: If criticized, engage respectfully with evidence
- Community testimonials: Real users sharing A1C improvements counter criticism
- Medical disclaimer always prominent

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\hline

\end{tabular}

\end{table}

Competitive Pricing Comparison:

- Noom: \$60/month (expensive, time-intensive)
- Dexcom Stelo CGM: \$99/month (hardware required)
- MyFitnessPal Premium: \$9.99/month (not prediabetes-specific)
- GlucoSnap positioning: Premium value at mid-tier price

9.3 Conversion Strategy

Paywall Placement:

- After 5th scan of the day: "You've reached your daily scan limit. Upgrade for unlimited scans."
- When viewing food sequencing (free users): Blurred card + "Unlock advice with Pro"
- After 7 days: "You've scanned 25 meals this week. See your full meal patterns with Pro."

- When viewing A1C roadmap: "Track your 90-day reversal journey with Pro"

Trial Strategy:

- 7-day free trial on annual plan
- No credit card required to start (reduces friction)
- In-app reminders at Day 5 of trial

Target Conversion Rates:

- Month 1: 6% free-to-paid
- Month 6: 8% free-to-paid
- Month 12: 10% free-to-paid (Cal AI achieves 8-12%[14])

9.4 Revenue Projections

Conservative Scenario (Year 1):

- 5,000 downloads from YouTube channel
- 8% paid conversion → 400 paying users
- \$9.99/month average → \$3,996 MRR = \$48K ARR

Moderate Scenario (Year 1):

- 20,000 downloads
- 10% conversion → 2,000 paying users
- \$9.99/month → \$19,980 MRR = \$240K ARR

Comparable: Cal AI (18-24 months)

- 5M downloads, \$2M MRR from 200K paying users[9][14]
- GlucoSnap niche is smaller but more targeted—higher conversion expected

9.5 Unit Economics

Cost Structure at Scale (10,000 users):

\begin{table}

Cost Category	Monthly Cost
OpenAI API (5 scans/day avg)	\$15,000
Hosting (Railway)	\$400
RevenueCat (1% revenue)	\$845
Total	\$16,245

Revenue Metric	Value
10,000 users at \$9.99/month	\$99,900 MRR
Minus costs	-\$16,245
Gross margin	\$83,655 (84%)

Table 24: Unit economics at 10K users with 10% conversion

Key Insight: Need to convert free users to paid. Cal AI does 8-12% conversion[14].

10. Regulatory and Compliance

10.1 Wellness Tool Positioning (Not Medical Device)

Critical Classification:

GlucoSnap is positioned as a **wellness and nutrition tool**, NOT a medical device. This avoids FDA regulation while still providing value.

Language Strategy:

- ✓ Use: "Educational tool," "Wellness companion," "Nutritional guidance"
- ✗ Avoid: "Diagnose," "Treat," "Medical advice," "Prescription"

Regulatory Risk Mitigation:

FDA regulates medical devices that diagnose or treat disease. Cal AI avoids this by calling itself "educational, not diagnostic." [15]
GlucoSnap uses same positioning.

Disclaimer Language (Visible in App):

"GlucoSnap provides educational information about food and blood sugar management. It is not a medical device and does not diagnose or treat any medical condition. Always consult your healthcare provider before making health decisions."

10.2 App Store Requirements

Apple App Store:

- Category: Health & Fitness
- Age Rating: 4+ (no restricted content)
- Privacy Nutrition Label: Health/Fitness data, User Content, Identifiers
- App Review: Expect health app scrutiny—have medical reviewer contacts ready

Google Play Store:

- Category: Health & Fitness
- Content Rating: Everyone
- Sensitive permissions justification: Camera (meal scanning), Notifications (post-meal reminders)

10.3 Privacy Policy and Terms of Service

Required Disclosures:

- What data is collected: Photos, A1C values, meal logs, device identifiers
- How data is used: AI analysis, progress tracking, personalization
- Data sharing: None to third parties for marketing; OpenAI for AI processing only
- Data retention: Photos 90 days, nutrition data indefinite, user can delete anytime
- User rights: GDPR compliance (EU users), CCPA compliance (CA users)

HIPAA Considerations:

GlucoseSnap is NOT a HIPAA-covered entity (not healthcare provider or insurer). Standard data protection practices apply, but HIPAA compliance not required.

10.4 Medical Disclaimer

Prominent Disclaimer (Multiple Locations):

1. App onboarding
2. Settings → About section
3. App Store description
4. Website footer

Exact Language:

"GlucoSnap is an educational wellness tool designed to provide nutritional information about Glycemic Load and food choices. It is not intended to diagnose, treat, cure, or prevent any disease or medical condition. The information provided by GlucoSnap should not replace professional medical advice. Always consult with your physician or qualified healthcare provider before making any changes to your diet, exercise routine, or health management plan. Individual results may vary."

11. Launch Strategy

11.1 Pre-Launch Activities (Weeks -4 to 0)

Week -4: Closed Beta

- 50 beta testers recruited from YouTube community
- TestFlight (iOS) and Google Play Internal Testing (Android)
- Collect feedback on UX, scan accuracy, feature requests
- Fix critical bugs, optimize performance

Week -3: Content Creation

- Record 5 YouTube videos showcasing app features
- Create App Store screenshots and preview video
- Write blog post: "Why I Built GlucoSnap"
- Prepare social media assets (Instagram, Twitter, Reddit)

Week -2: Marketing Preparation

- Finalize App Store listing (title, description, keywords)
- Set up RevenueCat products and pricing

- Configure analytics tracking (PostHog)
- Create landing page with email signup

Week -1: App Store Submission

- Submit to Apple App Review (expect 2-5 day review)
- Submit to Google Play Review (expect 1-3 day review)
- Prepare launch day announcement
- Schedule YouTube launch video

11.2 Launch Day (Day 0)

Launch Sequence:

1. **6 AM EST:** Apps go live on Apple App Store and Google Play Store
2. **8 AM EST:** Publish YouTube launch video: "I Built an App to Reverse Prediabetes"
3. **9 AM EST:** Email list announcement (from pre-launch signups)
4. **10 AM EST:** Reddit post in r/prediabetes (community announcement)
5. **12 PM EST:** Twitter/Instagram posts with app link
6. **2 PM EST:** Product Hunt launch (optional, additional visibility)

Launch Goals:

- 500 downloads Day 1
- 100 active users (completed first scan)
- 10 premium conversions
- 4.5+ App Store rating (from beta testers rating immediately)

11.3 Go-to-Market Strategy

Primary Distribution Channel: YouTube

- Existing prediabetes audience (established trust)
- Content strategy: "I scanned everything I ate for 7 days" video series
- Call-to-action in every video: "Download GlucoSnap (link in description)"
- Free user acquisition—no paid ads initially

Secondary Channels:

Reddit (r/prediabetes - 47K members):

- Authentic community engagement (not spammy)
- Share personal journey and app development story
- Respond to "what should I eat" posts with helpful advice + app mention

App Store Optimization (ASO):

- Title: "GlucoSnap: Prediabetes Reversal"
- Subtitle: "Photo Scan Meals, Track Glycemic Load"
- Keywords: prediabetes, blood sugar, glycemic load, A1C, diabetes prevention, meal scanner
- Category: Health & Fitness

Community Beta Testers as Advocates:

- 50 beta testers become word-of-mouth spreaders
- Incentive: Lifetime free premium access for testimonials
- Encourage social sharing of success stories

11.4 Post-Launch Growth (Months 1-12)

Month 1-3: Product-Market Fit Validation

- Goal: 2,000 downloads, 40% Day-7 retention, 6% conversion
- Weekly user interviews (10 users/week)
- Iterate on top user feedback (prioritize P1 features)
- YouTube content: Weekly app update videos

Month 4-6: Feature Expansion

- Launch P1 features (CGM integration, barcode scanner, meal history insights)
- Goal: 10,000 downloads, 8% conversion
- Begin light paid acquisition testing (Facebook/Instagram ads, small budget)

Month 7-12: Scale Growth

- Goal: 25,000 downloads, 10% conversion, \$35K MRR
- Launch P2 features (community, AI chat assistant)

- Expand content: Guest appearances on diabetes podcasts
- Press outreach: TechCrunch health tech beat, diabetes blogs

11.5 Competitive Advantage in Distribution

Biggest Advantage: YouTube Channel

Cal AI grew through viral TikTok/YouTube content of people scanning meals[9]. GlucoSnap has existing prediabetes audience. Distribution flywheel:

1. Post scan results on YouTube: "I scanned everything I ate for 7 days"
2. Community beta testers become word-of-mouth spreaders
3. Reddit pain points analyzed = perfect marketing copy
4. Each video = free user acquisition, no paid ads needed initially

12. Development Roadmap

12.1 MVP Launch (Weeks 1-8)

Goal: Ship core value proposition with production-grade AI accuracy to beta testers.

Week	Milestone
1	Rust backend API scaffolding, PostgreSQL schema, auth system, Redis cache
2	OpenAI Vision API integration with master prompt, complexity classifier, photo storage (R2)
3	React Native/Expo project setup, camera screen with plate overlay, scan results UI
4	Daily GL dashboard, meal history screen, onboarding flow, confidence scoring
5	A1C Reversal Roadmap, food sequencing cards, swap engine, portion bias correction
6	RevenueCat integration, paywall, subscription management, dish name shortcut
7	Push notifications, post-meal walk reminders, streak system, editable portions
8	Beta testing (50 users), accuracy validation (50 real meals), bug fixes, App Store submission

Table 25: 8-week MVP development sprint with accuracy features

MVP Feature Set (P0 + Critical Accuracy Features):

Core Features (P0):

- Onboarding flow (5 screens)
- Core scan feature (photo → GL analysis)
- Daily GL budget tracker
- A1C reversal roadmap
- Food sequencing coach
- Safer swap engine
- Basic meal history
- Premium subscription paywall

Accuracy & Safety Features (P0 - Critical for Clinical Credibility):

- **Complexity classifier** (SIMPLE/COMPLEX_B/COMPLEX_C routing)
- **Master system prompt** (forced visual estimation, no serving size assumptions)

- **Conservative GL fallback** (safety net for unidentified foods)
- **Plate diameter calibration overlay** (camera UI reference circle)
- **Systematic bias correction** (large portion adjustment)
- **Confidence scoring system** (LOW/MEDIUM/HIGH with GL ranges)
- **Dish name shortcut** (one-tap confirmation for complex dishes)
- **Editable portion confirmation** (human-in-the-loop correction)
- **Redis pHash caching** (7-day cache, instant results for repeated meals)

Implementation Priority for Accuracy Features:

Feature	Effort	Impact	Implement When
Complexity Classifier	2 hours	Very High	Day 1 of build
Master System Prompt	1 hour	Very High	Day 1 of build
Conservative GL Fallback	3 hours	Very High	Day 1 of build
Plate Calibration Overlay	1 day	High	Week 2
Systematic Bias Correction	4 hours	High	Week 2
Confidence Scoring	1 day	High	Week 2
Editable Portion Confirmation	2 days	Medium-High	Week 3
Dish Name Shortcut	2 days	High	Week 3
Two-Pass Architecture	3 days	Very High	Week 4

Table 26: Accuracy feature implementation priority

Critical Implementation Note:

The three P0 accuracy features (complexity classifier, master prompt, conservative fallback) can be implemented in under one day and will prevent the most dangerous failure modes (false "safe" signals on high-GL opaque foods) before any mobile UI code is written.

12.2 Version 1.1 (Weeks 9-16)

Goal: Feature parity with competitors + unique differentiators.

Feature	Priority
Barcode scanner (packaged foods)	P1
Weekly auto-generated insights	P1
Educational "Learn Hub" (20+ articles)	P1
Home screen widget (iOS/Android)	P1
Doctor-ready PDF export	P1
CGM integration (Terra API)	P1

Table 27: Version 1.1 feature additions

12.3 Version 1.2+ (Weeks 17-24)

Goal: Advanced features for retention and virality.

Feature	Priority
Community features (success stories, meal share)	P2
Accountability buddy matching	P2
Vegetarian-specific expanded swap library	P2
Menopause mode (targeted advice track)	P2
AI chat assistant (RAG-based)	P2
Apple Watch / Wear OS companion	P2

Table 28: Version 1.2 advanced features

12.4 Technical Debt and Optimization

Ongoing (Parallel to Feature Development):

- Performance optimization (reduce scan time to <3 seconds)
- AI prompt refinement (improve food recognition accuracy)
- Database query optimization (meal history loads <1 sec)
- Error monitoring and crash reduction (99.9% crash-free rate)
- Accessibility improvements (WCAG 2.1 AA compliance)
- Test coverage (80%+ backend, 70%+ frontend)

13. Success Metrics and Analytics

13.1 Key Performance Indicators (KPIs)

Acquisition Metrics:

- Total downloads (App Store + Google Play)
- Download source attribution (YouTube, Reddit, ASO, paid ads)
- Cost per install (CPI) - if running paid ads
- App Store ranking (Health & Fitness category)

Activation Metrics:

- Onboarding completion rate
- Time to first scan
- First scan success rate (result delivered <5 sec)
- Day 1 retention

Engagement Metrics:

- Daily Active Users (DAU)
- Monthly Active Users (MAU)
- DAU/MAU ratio (stickiness)
- Scans per active user per day
- Median session duration
- Day 7, Day 30, Day 90 retention

Monetization Metrics:

- Free-to-paid conversion rate
- Trial-to-paid conversion rate
- Monthly Recurring Revenue (MRR)
- Average Revenue Per User (ARPU)
- Customer Lifetime Value (LTV)
- Churn rate

Health Outcome Metrics:

- Average A1C improvement (90-day cohorts)
- Percentage of users achieving A1C <5.7
- Average daily GL adherence (staying under budget)

- Streak achievement rate (7-day, 30-day, 90-day)

Satisfaction Metrics:

- Net Promoter Score (NPS)
- App Store rating (iOS + Android average)
- Customer support ticket volume
- Feature request frequency

13.2 Analytics Implementation

Event Tracking (PostHog):

Critical Events:

- app_opened
- onboarding_started
- onboarding_completed
- first_scan_completed
- scan_initiated
- scan_result_viewed
- advice_card_expanded
- swap_suggestion_accepted
- meal_logged
- daily_gl_budget_exceeded
- streak_achieved (with milestone)
- a1c_value_entered
- paywall_viewed
- trial_started
- subscription_purchased
- subscription_cancelled

User Properties:

- a1c_baseline
- a1c_goal
- dietary_profile
- subscription_tier
- days_since_signup
- total_scans
- current_streak

- avg_daily_gl

13.3 Experiment Framework

A/B Testing Opportunities:

- Onboarding flow variations (3-screen vs 5-screen)
- Paywall messaging and placement
- Pricing tiers (\$9.99 vs \$12.99 monthly)
- Free tier scan limits (3 vs 5 vs unlimited with ads)
- Notification timing and copy
- Swap suggestion presentation (cards vs list)

Testing Methodology:

- Minimum sample size: 1,000 users per variant
- Statistical significance threshold: 95% confidence
- Duration: Minimum 7 days per experiment
- Tools: PostHog built-in experimentation + RevenueCat experiments

14. Risk Management

14.1 Identified Risks and Mitigation Strategies

Risk 1: FDA Regulatory Risk

- **Likelihood:** Low
- **Impact:** Critical
- **Description:** FDA classifies app as medical device, requires clearance
- **Mitigation:** Cal AI avoids this by calling itself "educational, not diagnostic." [15] Use same positioning. Clear disclaimers throughout app. Never use diagnostic language.
- **Contingency:** Consult health regulatory attorney pre-launch. Have medical review board review all app content.

Risk 0: AI Accuracy and User Safety (NEW - HIGHEST PRIORITY)

- **Likelihood:** High (inherent to technology)
- **Impact:** Critical (user trust, clinical outcomes, liability)

- **Description:** AI meal recognition inaccuracies lead to incorrect GL estimates, causing users to (a) skip needed post-meal walks for underestimated high-GL meals, or (b) unnecessarily restrict safe foods due to overestimation
- **Research Evidence:**
 - Standard GPT-4o: 59% food recognition accuracy, 47.9% carbohydrate MAPE[16][18]
 - Custom prompt engineering: Improves to 74% recognition, 15-20% carb accuracy[16]
 - Systematic underestimation bias for large portions (slope -0.33 to -0.50)[18]
 - Hidden ingredient failures: Opaque foods (soups, smoothies) have zero visual information[20]
- **Mitigation Strategy - Multi-Layer Safety Architecture:**
 - **Layer 1: Complexity Classifier** - Route opaque foods to user input modal, not blind AI guess
 - **Layer 2: Conservative Bias** - For prediabetes, overestimation safer than underestimation (triggers walk vs. causes spike)
 - **Layer 3: Confidence Scoring** - Display GL as range (20-40) when confidence LOW, not false-precision point estimate
 - **Layer 4: Human Oversight** - Editable portion confirmation, dish name input for complex meals
 - **Layer 5: Fallback Ranges** - Pre-defined conservative GL ranges for unidentifiable foods
 - **Layer 6: Educational Framing** - "This is an estimate based on visual analysis" disclaimer on every scan
- **Validation Protocol:**
 - Beta testing: 50 users scan 500+ meals (10 meals each over 1 week)
 - Accuracy audit: Compare AI estimates to manual dietitian analysis for 100 reference meals
 - Safety threshold: 85%+ correct Safe/Moderate/High classification (not precise macro accuracy)
 - Error analysis: Document and categorize all misclassifications
 - Continuous improvement: User corrections feed fine-tuning dataset
- **Contingency Plans:**

- If classification accuracy <80%: Implement two-pass architecture immediately (Week 4 roadmap)
- If dangerous underestimation pattern detected: Increase conservative bias multiplier
- If user trust issues: Add dietitian review option (premium feature)
- If liability concerns: Strengthen disclaimers, add "Not medical advice" banner on every scan
- **Success Metrics:**
 - Spike risk classification accuracy: >85% (primary safety metric)
 - User-reported "felt accurate" rating: >8/10
 - False negative rate (High GL meal flagged as Safe): <5%
 - User corrections per scan: <15% (indicates AI baseline quality)

Critical Safety Principle:

For prediabetes management, it is always safer to overestimate GL than underestimate. A false HIGH warning triggers a post-meal walk (harmless, possibly beneficial even if meal was actually moderate). A false SAFE signal for a high-GL dish causes blood sugar spike (harmful, undermines reversal efforts).

All architectural decisions prioritize minimizing false negatives over false positives.

Risk 2: OpenAI API Costs Scaling

- **Likelihood:** Medium
- **Impact:** High
- **Description:** At scale, OpenAI API costs become prohibitive
- **Mitigation:**
 - Implement aggressive image compression before sending to API
 - Cache results for identical foods (reduce repeat API calls)
 - Consider training custom lightweight model for common foods (P2)
 - Implement rate limiting for free tier users

- **Contingency:** Increase premium subscription price to \$14.99/month to maintain margin.

Risk 3: App Store Rejection

- **Likelihood:** Medium
- **Impact:** High
- **Description:** Health apps face strict App Store scrutiny
- **Mitigation:**
 - Clear privacy policy and medical disclaimer before submission
 - Comprehensive data usage disclosures
 - No false medical claims in copy
 - Have medical reviewer validate all content
- **Contingency:** Web-based Progressive Web App (PWA) version if App Store approval delayed.

Risk 4: Competition - GlycoAI, Glycemic Snap

- **Likelihood:** High
- **Impact:** Medium
- **Description:** Existing apps add prediabetes-specific features
- **Mitigation:**
 - Differentiate through full prediabetes reversal system (A1C tracking, reversal roadmap), not just GI scanning
 - Leverage YouTube distribution advantage
 - Build community moat through YouTube audience trust
- **Contingency:** Focus on premium features (CGM integration, community, AI chat) that require significant R&D.

Risk 5: Low User Conversion (Free to Paid)

- **Likelihood:** Medium
- **Impact:** High
- **Description:** Users don't see enough value to pay \$12.99/month
- **Mitigation:**
 - Aggressive paywall experimentation (timing, messaging, pricing)
 - Demonstrate clear ROI: "Users who track with Pro reduce A1C 2× faster"

- Offer annual plan (\$79.99/year) for better LTV
- Consider lower price point (\$7.99/month) if conversion remains <5%
- **Contingency:** Introduce ad-supported free tier (non-intrusive banner ads).

Risk 6: AI Meal Recognition Accuracy Issues

- **Likelihood:** Medium
- **Impact:** Medium
- **Description:** GPT-4o Vision misidentifies foods, provides inaccurate GL calculations
- **Mitigation:**
 - Cross-reference AI results with USDA FoodData Central for validation
 - Allow user corrections and feedback loop for AI improvement
 - Clear messaging: "This is an estimate based on visual analysis"
 - Manual entry option for ambiguous foods
- **Contingency:** Implement human review for first 500 scans to validate accuracy threshold.

14.2 Risk Monitoring and Response

Weekly Risk Review:

- Monitor KPIs against targets
- Track customer support ticket themes
- Review App Store reviews for sentiment
- Assess API cost burn rate vs projections
- Evaluate conversion funnel drop-off points

Escalation Criteria:

- Day 7 retention <25% → Immediate UX investigation
- Conversion rate <4% after Month 1 → Pricing/value prop adjustment
- App Store rating <4.0 → Emergency bug fixes and user outreach
- API costs >30% of MRR → Cost optimization sprint

15. Appendices

15.1 Glossary of Terms

Term	Definition
A1C	Hemoglobin A1C test measuring average blood glucose over 3 months
CGM	Continuous Glucose Monitor - wearable device tracking real-time glucose
GL	Glycemic Load - measure of blood sugar impact of a food portion
GI	Glycemic Index - relative ranking of carbohydrate foods (0-100 scale)
Prediabetes	A1C 5.7-6.4%, elevated blood sugar not yet diabetic
Spike Risk	Classification of meal's blood sugar impact (Low/Moderate/High)
DAU	Daily Active Users
MAU	Monthly Active Users
MRR	Monthly Recurring Revenue
ARPU	Average Revenue Per User
LTV	Customer Lifetime Value

Table 29: Key terminology and definitions

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15.3 Revision History

Version	Date	Author	Changes
0.1	Feb 20, 2026	Product Team	Initial draft
0.5	Feb 24, 2026	Product Team	Added technical architecture
1.0	Feb 26, 2026	Product Team	Complete PRD ready for development

Table 30: Document revision history

Document Approval

Approved for Development:

Product Owner
Date: _____

Engineering Lead
Date: _____

Design Lead
Date: _____

END OF DOCUMENT

GlucoSnap Product Requirements Document v1.0
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